

SERVICE DATA
FILE NO. 300-450

TOSHIBA

AIR-CONDITIONER SPLIT (CEILING TYPE) HEAT PUMP

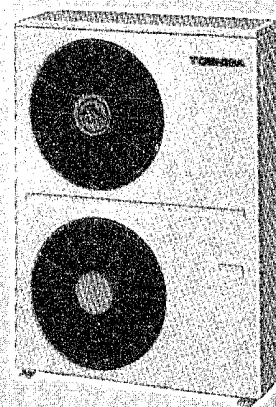
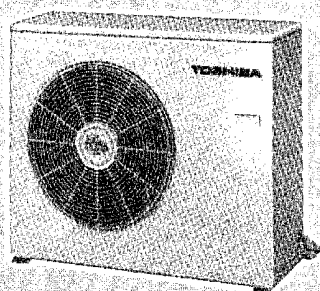
RAV-450CHJE/S450HJE

RAV-560CHJE/S560HJE

RAV-710CHE8/S710HE8

RAV-1000CHE8/S1000HE8

RAV-1250CHE8/S1250HE8



RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000HE8
RAV-1250CHE8/S1250HE8

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RAV-450CHJE/S450HJI
RAV-560CHJE/S560HJI
RAV-710CHE8/S710HE
RAV-1000CHE8/S1000I
RAV-1250CHE8/S1250I

1. SPECIFICATIONS

ITEM		MODEL	RAV-450CHJE		RAV-560CHJE	
Cooling capacity		k cal/h	4,000		5,000	
	*1	BTU/h	16,000		20,000	
		kW	4.7		5.8	
Heating capacity		k cal/h	4,500		5,250	
	*2	BTU/h	18,000		21,000	
		kW	5.2		6.1	
Power source		Phase	1		1	
		V	220/240		220/240	
		Hz	50		50	
Power consumption			COOLING	HEATING	COOLING	HEATING
		kW	2.07	1.93	2.1	1.95
Power factor		%	92.8	91.2	91.3	91.2
Running current		A	9.7	9.2	10.0	9.3
Starting current		A	51	51	51	51
Refrigerant	Name of refrigerant		R-22		R-22	
	Charge volume		1.15		1.6	
Refrigerant control			Capillary tube & Expansion valve		Capillary tube & Expansion valve	
Interconnection pipe	Gas side size	mm (in)	φ12.7 (1/2")		φ12.7 (1/2")	
	Coupler style		Flare		Flare	
	Liquid side size	mm (in)	φ6.4 (1/4")		φ6.4 (1/4")	
	Coupler style		Flare		Flare	
	Maximum length (of one way)	*3 m (ft)	25 (82')		25 (82')	
	Maximum height					
	Indoor unit higher	m (ft)	15 (49')		15 (49')	
	Outdoor unit higher	m (ft)	15 (49')		15 (49')	
Condensate drain pipe	Diameter (inside)	mm	φ20		φ20	
INDOOR UNIT Model			RAV-450CHJE		RAV-560CHJE	
Dimensions	Height	mm (in)	250 (9-7/8")		250 (9-7/8")	
	Width	mm (in)	900 (2'11-7/16")		1,200 (3'11-1/4")	
	Depth	mm (in)	710 (2'3-15/16")		710 (2'3-15/16")	
Net weight		kg (lbs)	30 (66)		39 (86)	
Heat exchanger type			Finned tybe		Finned tube	
Indoor fan type			Centrifugal fan		Centrifugal fan	
Air volume	High fan	m ³ /h (CFM)	930 (547)		1,320 (777)	
	Low fan	m ³ /h (CFM)	690 (406)		970 (571)	
Fan motor output		W	45		70	
Air filter			Washable		Washable	
OUTDOOR UNIT Model			RAV-S450HJE		RAV-S560HJE	
Dimensions	Height	mm (in)	740 (2'5-1/8")		740 (2'5-1/8")	
	Width	mm (in)	880 (2'10-5/8")		880 (2'10-5/8")	
	Depth	mm (in)	310 (1'3/16")		310 (1'3/16")	
Net weight		kg (lbs)	61 (134)		68 (150)	
Heat exchanger type			Finned tube		Finned tube	
Outdoor fan type			Propeller fan		Propeller fan	
Fan motor output		W	39		39	
Compressor	Model		PH207JA-4		PH207JA-4	
	Output	kW	1.5		1.5	
Safety device			High pressure switch Inner overload relay		High pressure switch Inner overload relay	
Auto louver (Motor kit)			Option (RBC-M2CE)		Option (RBC-M2CE)	

Specifications are subject to change without notice.

Note 1: Cooling Capacity is based on the following temperature conditions.

Room air temperature: 27°C DB (80°F DB)
19.5°C WB (67°F WB)
Ambient air temperature: 35°C DB (95°F DB)

Note 2: Heating Capacity is based on the following temperature conditions.

Room air temperature: 21°C DB (70°F DB)
7°C DB (45°F DB)
6°C WB (43°F WB)

Note 3: These mean equivalent length.

Note 4: Operating range of the units

Room air temperature
When cooling
Maximum 32°C DB, 22.5°C WB (90°F DB, 73°F WB)
Minimum 21°C DB, 15.5°C WB (70°F DB, 60°F WB)
When heating
Maximum 28°C DB (82°F DB)
Minimum 17°C DB (63°F DB)
Ambient air temperature
When cooling
Maximum 43°C DB (109°F DB)
Minimum 21°C DB (70°F DB)
When heating
Maximum 21°C DB (70°F DB)

RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000HE8
RAV-1250CHE8/S1250HE8

MODEL			RAV-710CHE8		RAV-1000CHE8		RAV-1250CHE8	
ITEM								
Cooling capacity	*1	k cal/h	6,300		9,000		11,600	
		BTU/h	25,200		36,000		46,400	
		kW	7.3		10.5		13.5	
Heating capacity	*2	k cal/h	6,800		9,300		11,900	
		BTU/h	27,200		37,200		47,600	
		kW	7.9		10.8		13.8	
Power source		Phase	3		3		3	
		V	380/415		380/415		380/415	
		Hz	50		50		50	
			COOLING	HEATING	COOLING	HEATING	COOLING	HEATING
Power consumption		kW	2.8	2.6	3.7	3.4	4.8	4.5
Power factor		%	86.0	85.3	86.1	86.1	86.6	85.5
Running current		A	4.7	4.4	6.2	5.7	8.0	7.6
Starting current		A	28	28	42	42	50	50
Refrigerant	Name of refrigerant		R-22		R-22		R-22	
	Charge volume	kg	2.0		2.7		3.5	
Refrigerant control			Capillary tube & Expansion valve		Capillary tube & Expansion valve		Capillary tube & Expansion valve	
Interconnection pipe	Gas side size	mm (in)	φ15.9 (5/8")		φ19 (3/4")		φ19 (3/4")	
	Coupler style		Flare		Flare		Flare	
	Liquid side size	mm (in)	φ6.4 (1/4")		φ9.5 (3/8")		φ9.5 (3/8")	
	Coupler style		Flare		Flare		Flare	
	Maximum length (of one way)	*3 m (ft)	25 (82')		25 (82')		25 (82')	
	Maximum height							
	Indoor unit higher	m (ft)	15 (49')		15 (49')		15 (49')	
	Outdoor unit higher	m (ft)	15 (49')		15 (49')		15 (49')	
Condensate drain pipe	Diameter (inside)	mm	φ20		φ20		φ20	
INDOOR UNIT Model			RAV-710CHE8		RAV-1000CHE8		RAV-1250CHE8	
Dimensions	Height	mm (in)	250 (9-7/8")		250 (9-7/8")		250 (9-7/8")	
	Width	mm (in)	1,200 (3'11-1/4")		1,350 (4'5-1/8")		1,650 (5'5")	
	Depth	mm (in)	710 (2'3-15/16")		710 (2'3-15/16")		710 (2'3-15/16")	
Net weight		kg (lbs)	42 (93)		43 (95)		53 (117)	
Heat exchanger type			Finned tube		Finned tube		Finned tube	
Indoor fan type			Centrifugal fan		Centrifugal fan		Centrifugal fan	
Air volume	High fan	m ³ /h (CFM)	1,320 (777)		1,680 (989)		2,100 (1,236)	
	Low fan	m ³ /h (CFM)	970 (571)		1,240 (730)		1,550 (912)	
Fan motor output		W	70		82		140	
Air filter			Washable		Washable		Washable	
OUTDOOR UNIT Model			RAV-S710HE8		RAV-S1000HE8		RAV-S1250HE8	
Dimensions	Height	mm (in)	790 (2'7-1/8")		1,240 (4'13/16")		1,240 (4'13/16")	
	Width	mm (in)	880 (2'10-5/8")		930 (3'5/8")		930 (3'5/8")	
	Depth	mm (in)	310 (1'3/16")		385 (1'3-3/16")		385 (1'3-3/16")	
Net weight		kg (lbs)	78 (172)		107 (236)		115 (254)	
Heat exchanger type			Finned tube		Finned tube		Finned tube	
Outdoor fan type			Propeller fan		Propeller fan		Propeller fan	
Fan motor output		W	63		39 + 63		39 + 63	
Compressor	Model		YH277JA		YH406JA		YH506JA	
	Output	kW	2.0		3.0		3.75	
Safety device			High pressure switch over current relay		High pressure switch over current relay Inner overload relay		High pressure switch over current relay Inner overload relay	
Auto louver (Motor kit)			Option (RBC-M2CE)		Option (RBC-M2CE)		Option (RBC-M2CE)	

Specifications are subject to change without notice.

Note 1: Cooling Capacity is based on the following temperature conditions.

Room air temperature: 27°C DB (80°F DB)
19.5°C WB (67°F WB)
Ambient air temperature: 35°C DB (95°F DB)

Note 2: Heating Capacity is based on the following temperature conditions.

Room air temperature: 21°C DB (70°F DB)
7°C DB (45°F DB)
Ambient air temperature: 6°C WB (43°F WB)

Note 3: These mean equivalent length.

Note 4: Operating range of the units

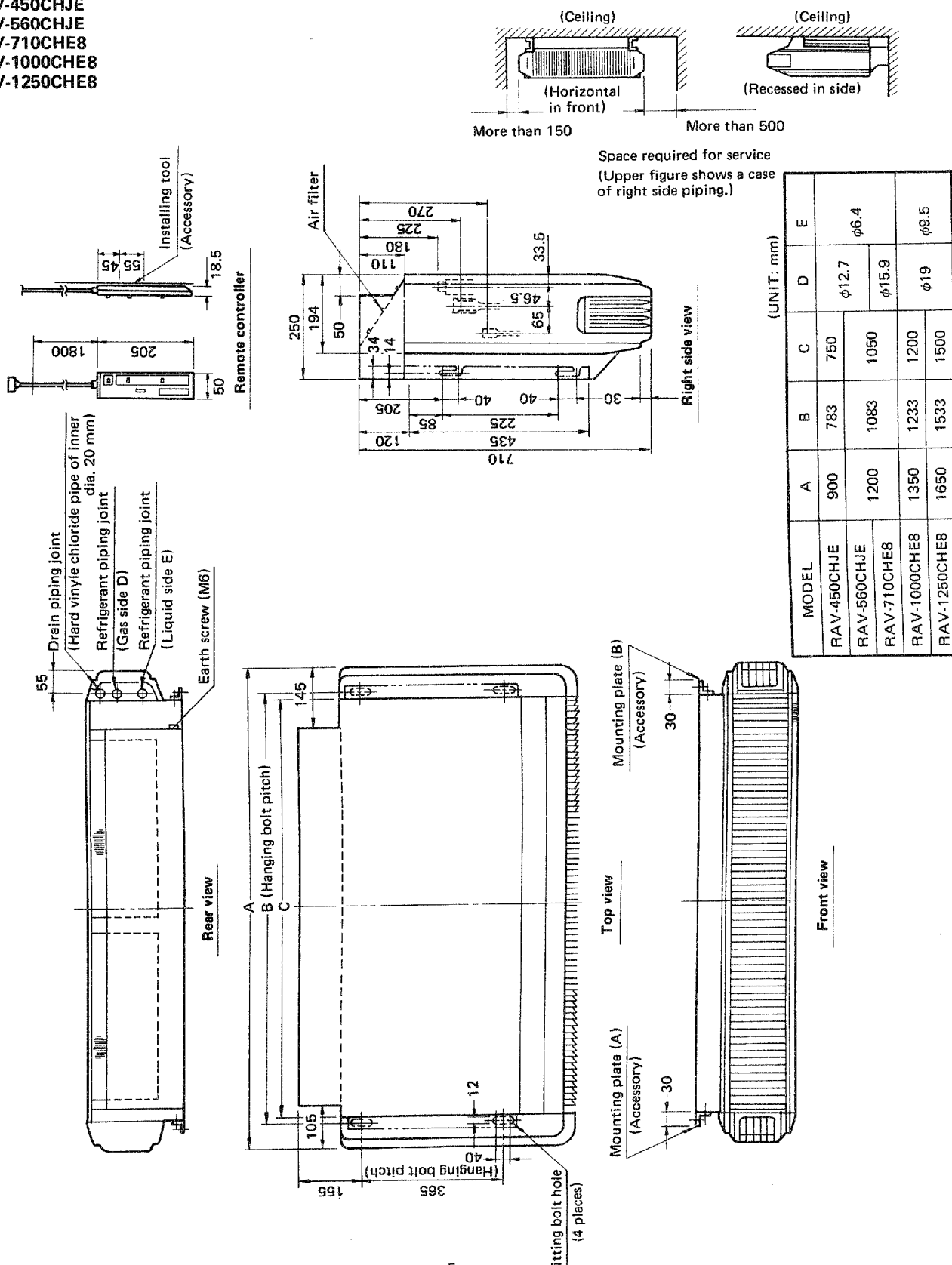
Room air temperature
When cooling
Maximum 32°C DB, 22.5°C WB (90°F DB, 73°F WB)
Minimum 21°C DB, 15.5°C WB (70°F DB, 60°F WB)
When heating
Maximum 28°C DB (82°F DB)
Minimum 17°C DB (63°F DB)
Ambient air temperature
When cooling
Maximum 43°C DB (109°F DB)
Minimum 21°C DB (70°F DB)

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000H
 RAV-1250CHE8/S1250H

2. CONSTRUCTION VIEWS

2-1. Indoor unit

RAV-450CHJE
 RAV-560CHJE
 RAV-710CHE8
 RAV-1000CHE8
 RAV-1250CHE8

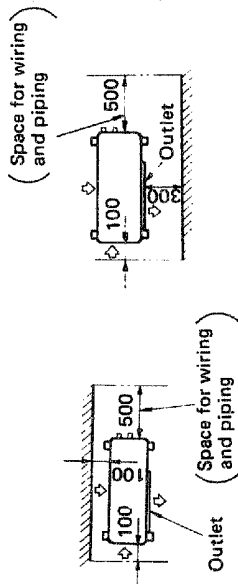
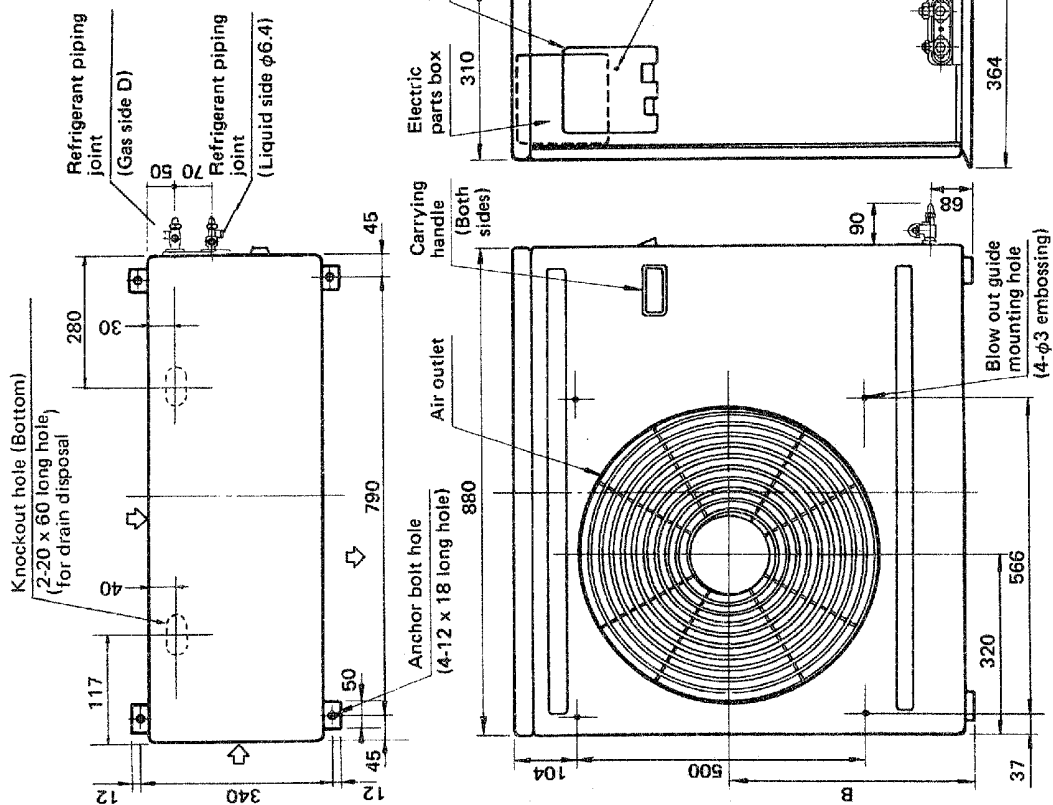


RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

2-2. Outdoor unit

RAV-S450HJE
 RAV-S560HJE
 RAV-S710HE8

Space required for service



When installed with the inlet faced to the wall side

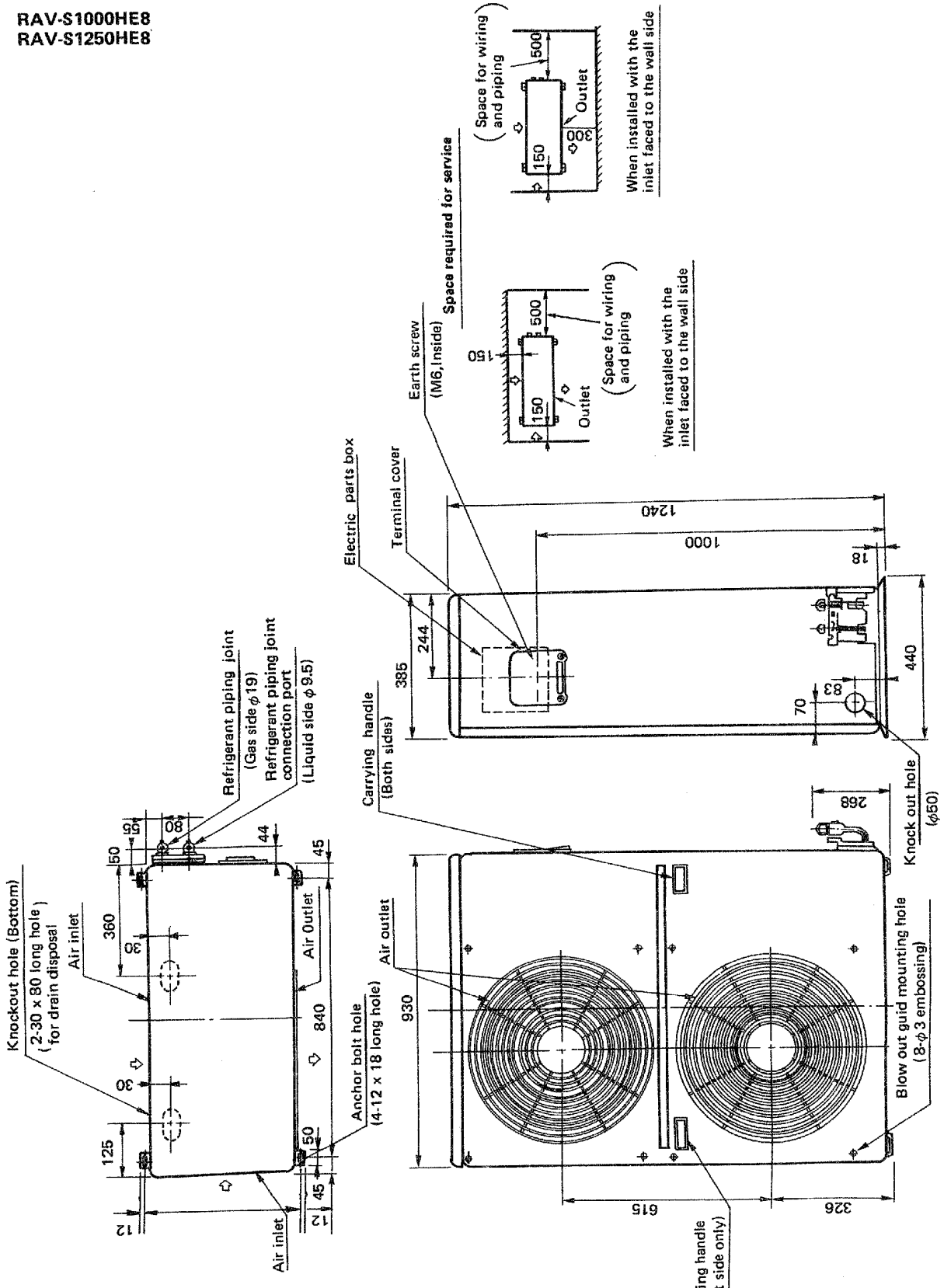
When installed with the inlet faced to the wall side

(Unit: mm)

MODEL	A	B	C
RAV-S450HJE	740	380	φ12.7
RAV-S560HJE	790	430	φ15.9

RAV-S1000HE8
RAV-S1250HE8

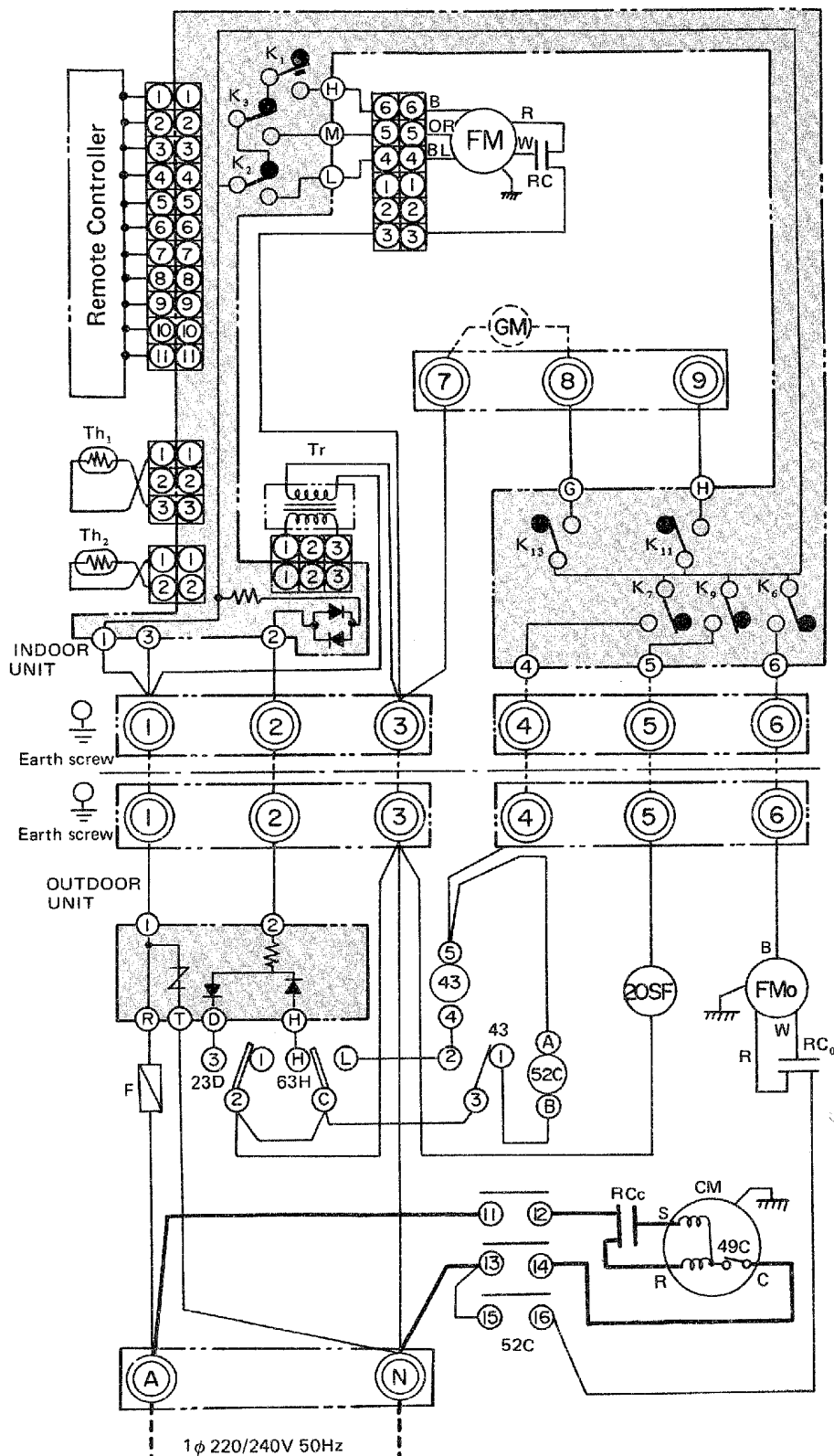
RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000H
RAV-1250CHE8/S1250H



RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000HE8
RAV-1250CHE8/S1250HE8

3. SCHEMATIC DIAGRAM

RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE



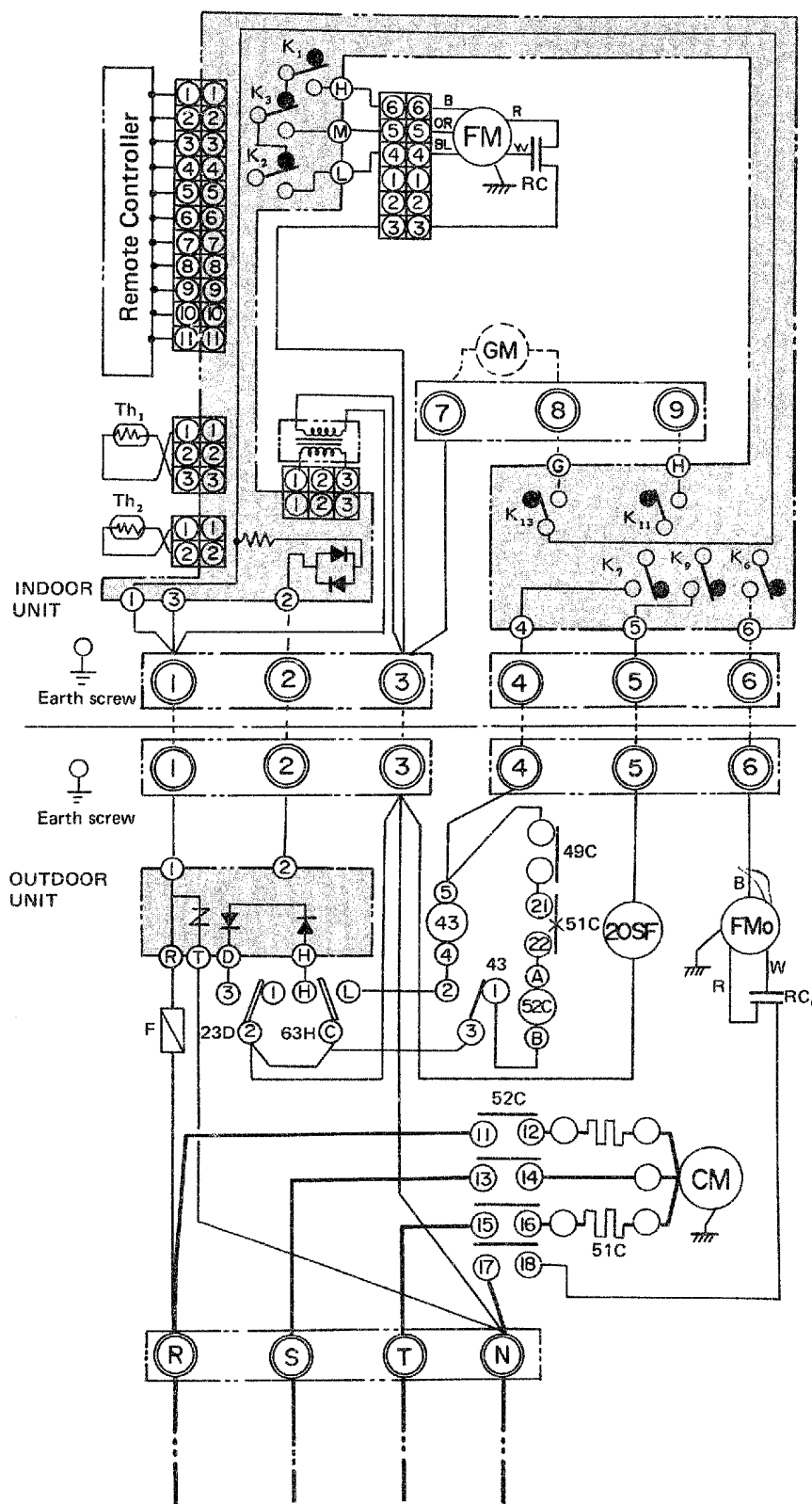
Mark	Description
CM	Compressor
FM	Fan Motor (Indoor)
FMo	Fan Motor (Outdoor)
GM	Geared Motor (Option)
52C	Magnetic Contactor
49C	Inner Overload Relay
63H	High-Pressure Switch
RCc	Running Capacitor (Compressor)
RC	Running Capacitor (Indoor)
RCo	Running Capacitor (Outdoor)
23D	Defrost Thermostat
20SF	Solenoid Coil
43	Relay
Tr	Transformer
Th ₁	Sensor for Heat-exchanger
Th ₂	Sensor for Room Temperature
K ₁ ~K ₁₃	Relay in print board
F	Fuse

Color Identification

B: Black
OR: Orange
BL: Blue
R: Red
W: White

RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHJE/S710HE
RAV-1000CHJE/S1000H
RAV-1250CHJE/S1250H

RAV-710CHJE/S710HE8



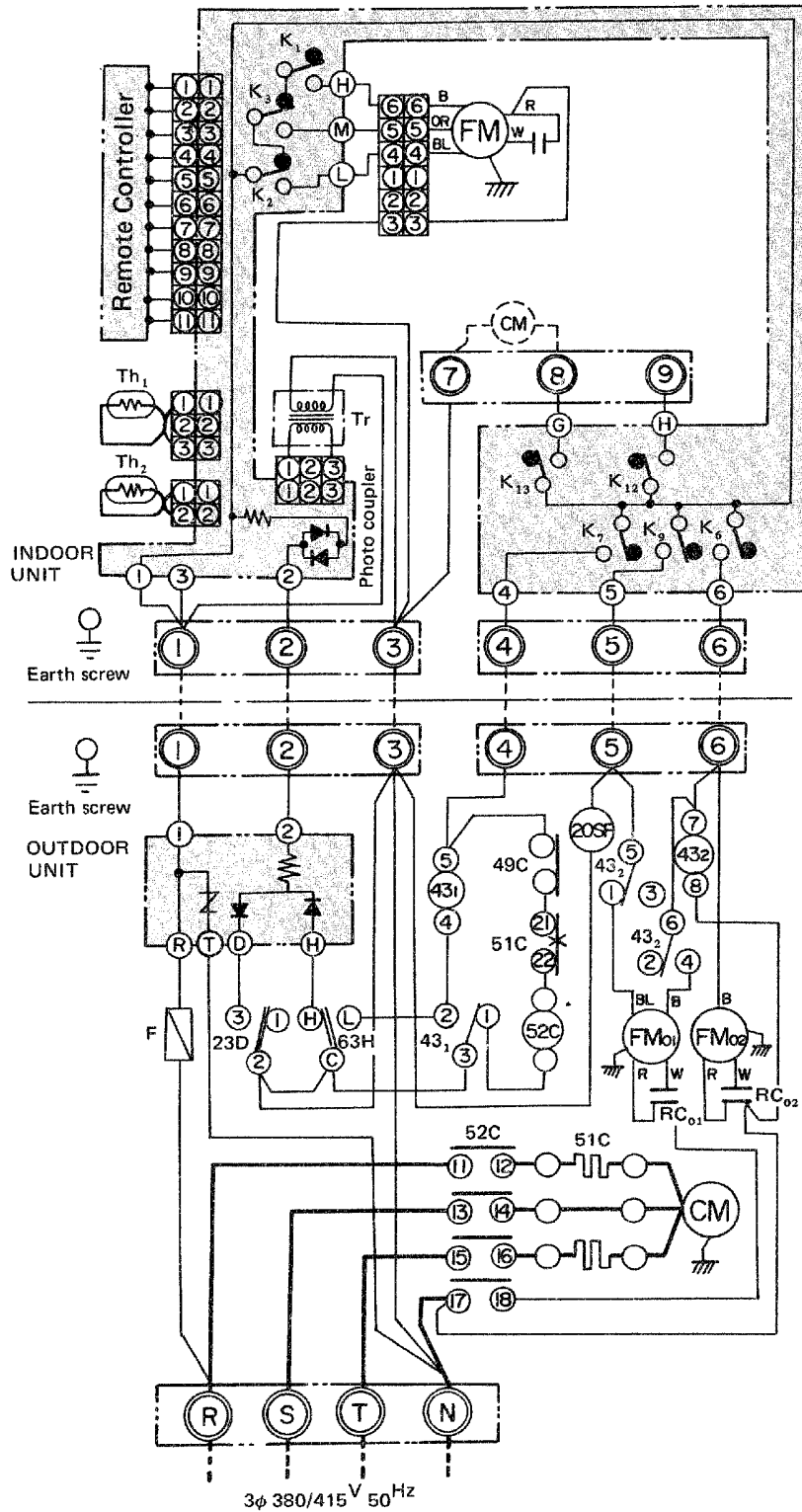
Mark	Description
CM	Compressor
FM	Fan Motor (Indoor)
FMo	Fan Motor (Outdoor)
GM	Geared Motor (Option)
52C	Magnetic Contactor
51C	Overcurrent Relay
49C	Thermostat for Compressor Case
63H	High-Pressure Switch
RC	Running Capacitor (Indoor)
RCo	Running Capacitor (Outdoor)
23D	Defrost Thermostat
20SF	Solenoid Coil
43	Relay
Tr	Transformer
Th ₁	Sensor for Heat-exchanger
Th ₂	Sensor for Room Temperature
K ₁ ~K ₁₃	Relay in print board
F	Fuse

Color Identification

B: Black
OR: Orange
BL: Blue
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RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000HE8
RAV-1250CHE8/S1250HE8

RAV-1000CHE8/S1000HE8
RAV-1250CHE8/S1250HE8



RC Wiring (1000CH)

Mark	Description
CM	Compressor
FM	Fan Motor (Indoor)
FM _{01,2}	Fan Motor (Outdoor)
GM	Geared Motor (Option)
52C	Magnetic Contactor
51C	Overcurrent Relay
49C	Inner Overload Relay
63H	High-Pressure Switch
RC	Running Capacitor (Indoor)
RC _{01,2}	Running Capacitor (Outdoor)
23D	Defrost Thermostat
20SF	Solenoid Coil
43 _{1,2}	Relay
Tr	Transformer
Th ₁	Sensor for Heat-exchanger
Th ₂	Sensor for Room Temperature
K ₁ ~ K ₁₃	Relay in print board
F	Fuse

Color Identification

B: Black
OR: Orange
BL: Blue
R: Red
W: White

RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000H
RAV-1250CHE8/S1250H

4. SPECIFICATIONS OF ELECTRICAL PARTS

4-1. RAV-450CHJE/S450HJE

No.	Parts name	Type	Specifications						
1	Compressor	PH207JA-4	Output (Rated) 1.5 kW, 2 pole, 1 phase, 220/240V, 50Hz						
			Winding resistance (Ω , at 20°C)		R-C		S-C		
					1.11		2.61		
2	Indoor unit Fan motor	AF-230-45-4E	Output (Rated) 45W, 4 pole, 1 phase, 220/240V, 50Hz						
			Winding resistance (Ω , at 20°C)		R-BR	B-BR	OR-BR	BL-BR	W-BR
					60.67	15.95	32.37	37.80	79.49
3	Outdoor unit Fan motor	AF-230-39K	Output (Rated) 39W, 6 pole, 1 phase, 220/240V, 50Hz						
			Winding resistance (Ω , at 20°C)		R-B		W-B		
					53.3		54.2		
4	Running capacitor for compressor	S45ALFPA506W	AC 450V, 50 μ F						
5	Running capacitor for indoor fan motor	40FAFN305UJ	AC 400V, 3 μ F						
6	Running capacitor for outdoor fan motor	35FAFN405UJ	AC 350V, 4 μ F						
7	Transformer	TB203	230V/12V						
8	Sensor for heat-exchanger temp.		Maximum input 1 mW	°C	10	20	30	40	50
				k Ω	9.9	6.6	4.5	3.1	2.2
9	Sensor for room temperature		Maximum input 450 mW	°C	10	15	20	25	30
				k Ω	20.7	16.1	12.6	10.0	8.0
10	High pressure switch	HTB-310	Tripping pressure 26 kg/cm ² G Resetting pressure 21 kg/cm ² G						
11	Defrost thermostat	A-605A	Tripping temperature -3°C Resetting temperature 8°C						
12	Solenoid coil	L27	AC 240V						
13	Relay	SA106H-230	AC 230V, 6A, 1ab						
14	Magnetic contactor	VC20FA	AC 240V, 20 A 3a						

4-2. RAV-560CHJE/S560HJE

Different points from above model are shown below and other specifications are the same as above.

No.	Parts name	Type	Specifications				
2	Indoor unit Fan motor	AF-230-70-4F	Output (Rated) 70W, 4 pole, 1 phase, 220/240V, 50Hz				
			Winding resistance (Ω , at 20°C)	R-B	OR-B	BL-B	W-B
				37.41	19.29	33.49	60.15
10	High pressure switch	HTB-T308	Tripping pressure 28 kg/cm ² G Resetting pressure 23 kg/cm ² G				

4-3. RAV-710CHE8/S710HE8

No.	Parts name	Type	Specifications						
1	Compressor	YH277JA	Output (Rated) 2.0kW, 2 pole, 3 phase, 380/415V, 50Hz						
			Winding resistance 4.43Ω, at 20°C						
2	Indoor unit Fan motor	AF-230-70-4F	Output (Rated) 70W, 4 pole, 1 phase, 220/240V, 50Hz						
			Winding resistance (Ω, at 20°C)		R-B	OR-B	BL-B	W-B	
3	Outdoor unit Fan motor	AF-230-63K-1	Output (Rated) 63W, 6 pole, 1 phase, 220/240V, 50Hz						
			Winding resistance (Ω, at 20°C)		R-B		W-B		
					40.0		43.2		
4	Running capacitor for indoor fan motor	40FAFN305UJ	AC 400V, 3μF						
5	Running capacitor for outdoor fan motor	35FAFN405UJ	AC 350V, 4μF						
6	Transformer	TB203	230V/12V						
7	Sensor for heat-exchanger temp.		Maximum input 1 mW	°C	10	20	30	40	50
				kΩ	9.9	6.6	4.5	3.1	2.2
8	Sensor for room temperature		Maximum input 450 mW	°C	10	15	20	25	30
				kΩ	20.7	16.1	12.6	10.0	8.0
9	High pressure switch	HTB-T308	Tripping pressure 28 kg/cm ² G Resetting pressure 23 kg/cm ² G						
10	Defrost thermostat	A-606	Tripping temperature -3°C Resetting temperature 11.5°C						
11	Solenoid coil	L27	AC 240V						
12	Relay	SA106H-230	AC 230V, 6A, 1ab						
13	Magnetic contactor	E4-10F-F							
14	Overcurrent relay	RC4-10S	Tripping current 7.5A Resetting manual						
15	Thermostat for compressor case	CS-7-TC1	Tripping temperature 105°C Resetting temperature 83°C						

4-4. RAV-1000CHE8/S1000HE8

4-5. RAV-1250CHE8/S1250HE8

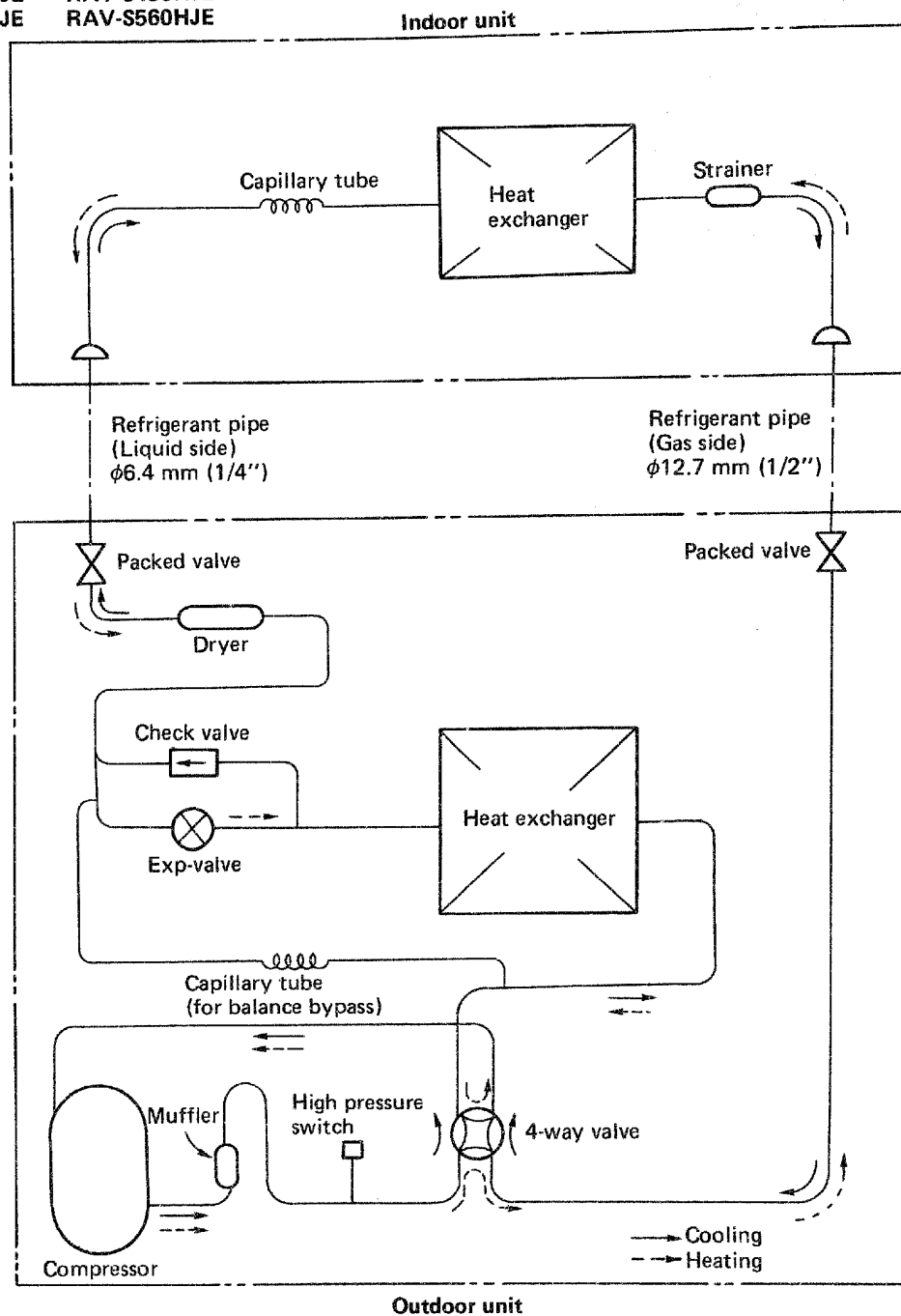
No.	Parts name	Type	Specifications				
1	Compressor	YH506JA	Output (Rated) 3.75 kW, 2 pole, 3 phase, 380/415V, 50Hz				
			Winding resistance 2.29Ω, at 20°C				
2	Indoor unit Fan motor	AF-230-140-4C	Output (Rated) 140W, 4 pole, 1 phase, 220/240V, 50Hz				
			Windint resistance (Ω, at 20°C)	R-B 27.8	OR-B 9.2	BL-B 20.2	W-B 24.6
4	Running capacitor for Indoor Fan motor	M40ALFL505W	AC 400V, 5μF				
13	Magnetic contactor	E4-25-F					
14	Overcurrent relay	RC4-35ER1	Tripping current 12A, Resetting manual				

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

5. REFRIGERANT PIPING DIAGRAM

INDOOR UNIT
 RAV-450CHJE
 RAV-560CHJE

OUTDOOR UNIT
 RAV-S450HJE
 RAV-S560HJE

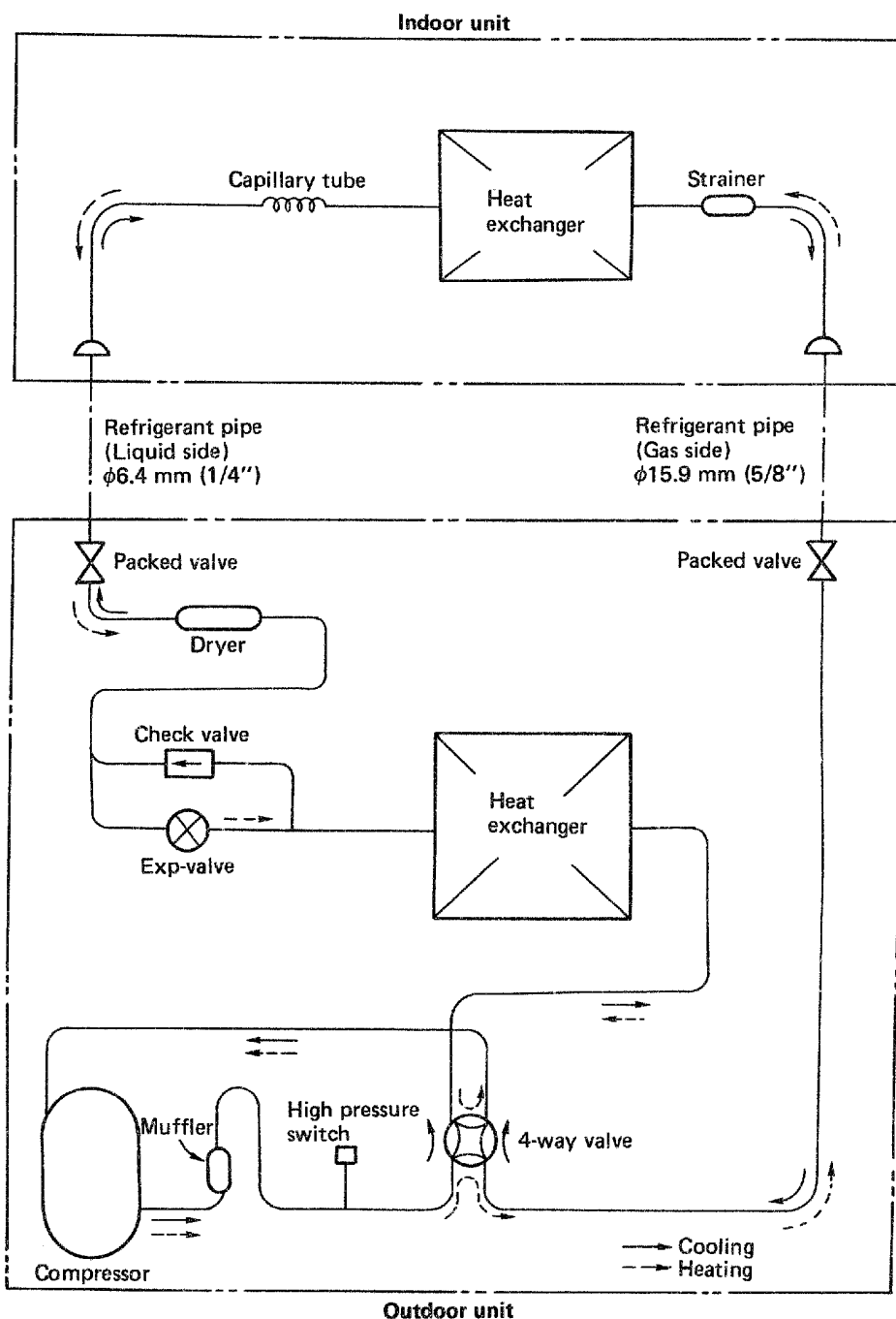


Model	Capillary tube
RAV-450CHJE	φ1.7 x 900ℓ x 2
RAV-560CHJE	φ1.5 x 1200ℓ x 4

Model	Check valve	Exp-valve	4-way valve
RAV-S450HJE	YCV5-3S-TF	BGX-2315DHS	V26-100B
RAV-S560HJE			

RAV-450CHJE/S450HJ
RAV-560CHJE/S560HJ
RAV-710CHE8/S710HE
RAV-1000CHE8/S1000
RAV-1250CHE8/S1250

INDOOR UNIT OUTDOOR UNIT
RAV-710CHE8 RAV-S710HE8



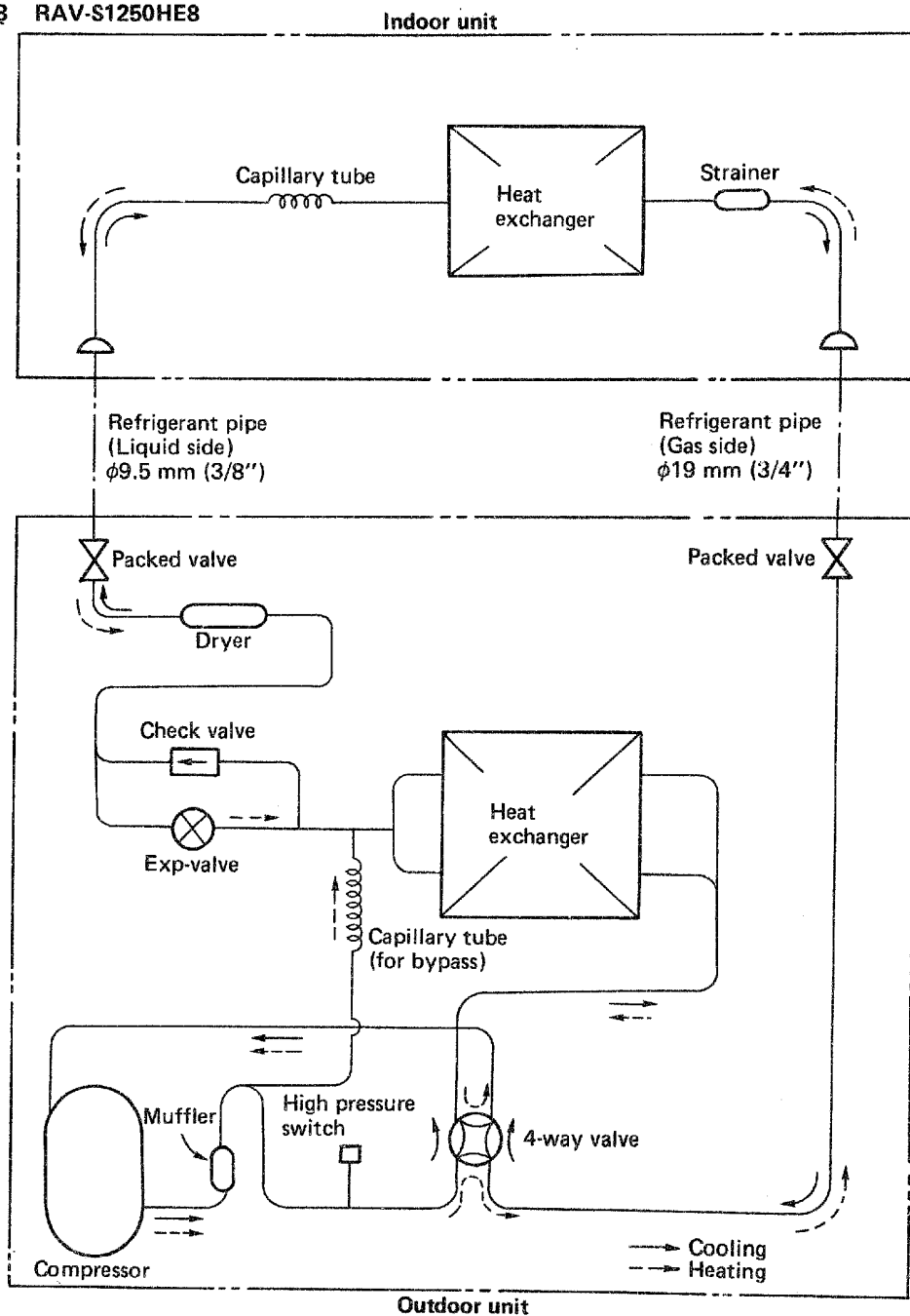
Model	Capillary tube
RAV-710CHE8	$\phi 1.7 \times 1100 \ell \times 3$

Model	Check valve	Exp-valve	4-way valve
RAV-S710HE8	YCV5-3S-TF	BGX-2320DHS	V26-100B

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

INDOOR UNIT
 RAV-1000CHE8
 RAV-1250CHE8

OUTDOOR UNIT
 RAV-S1000HE8
 RAV-S1250HE8

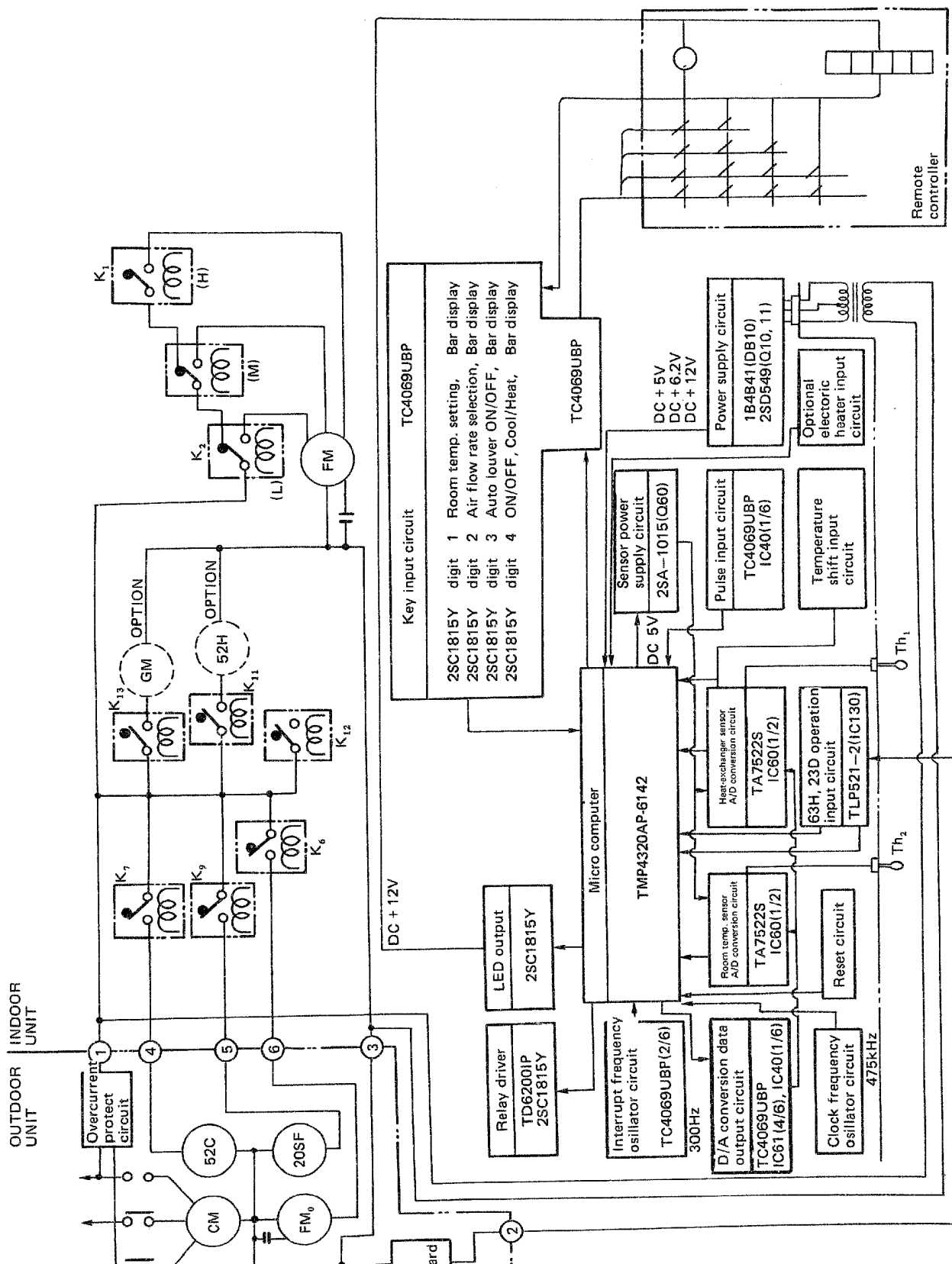


Model	Capillary tube
RAV-1000CHE8	$\phi 1.5 \times 900\ell \times 4$
RAV-1250CHE8	$\phi 1.5 \times 1100\ell \times 8$

Model	Check valve	Exp-valve	4-way valve	Capillary tube
RAV-S1000HE8	YCV5-3S-TF	BGX-2330DHS	V38-100B	$\phi 1.7 \times 1000\ell \times 1$
RAV-S1250HE8			V30-101	$\phi 2.0 \times 1000\ell \times 2$

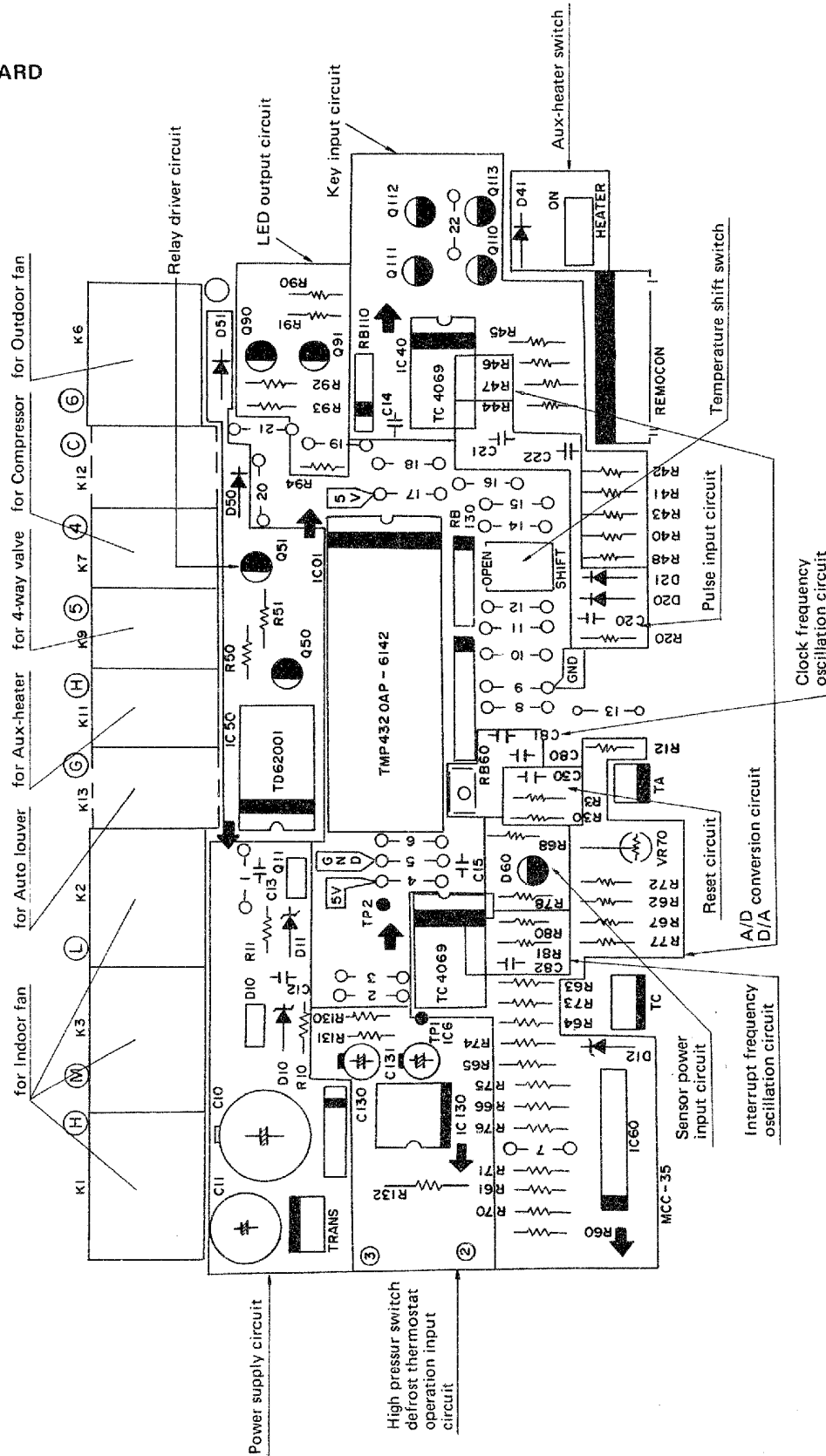
RAV-450CHJE/S450HJ
RAV-560CHJE/S560HJ
RAV-710CHE8/S710HE
RAV-1000CHE8/S1000
RAV-1250CHE8/S1250

6. CONTROL CIRCUIT BLOCK DIAGRAM



RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

INDOOR PC BOARD



RAV-450CHJE/S450HJ
RAV-560CHJE/S560HJ
RAV-710CHE8/S710HE
RAV-1000CHE8/S1000H
RAV-1250CHE8/S1250H

7. FUNCTION OF MICROCOMPUTER CONTROL

7-1. Room Temperature Control

- 1) Temperature adjustable range
Cooling: 21 ~ 32°C
Heating: 17 ~ 28°C
- 2) Differential: 1 deg(°C)
- 3) Timer for compressor restarting: 2.5 minutes
—compressor safety start— (1 minute at initial start-up)

NOTE: Regardless of the SELECTOR setting, the micro-computer controls the indoor fan motor as follows:

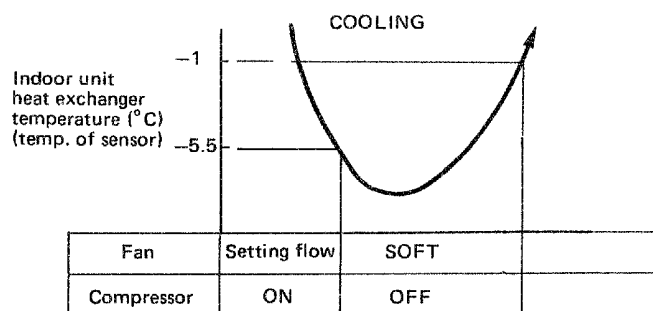
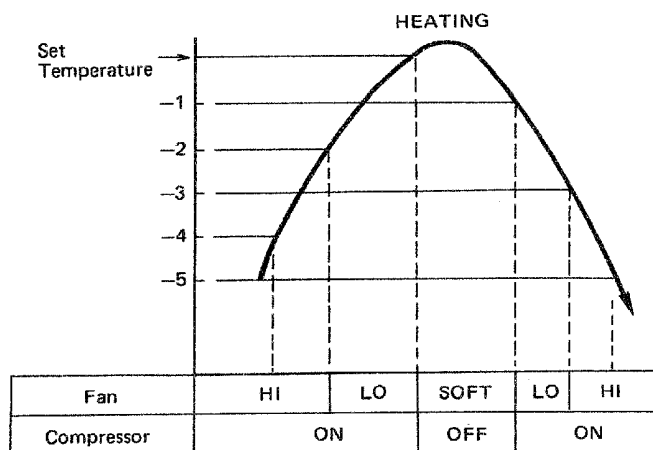
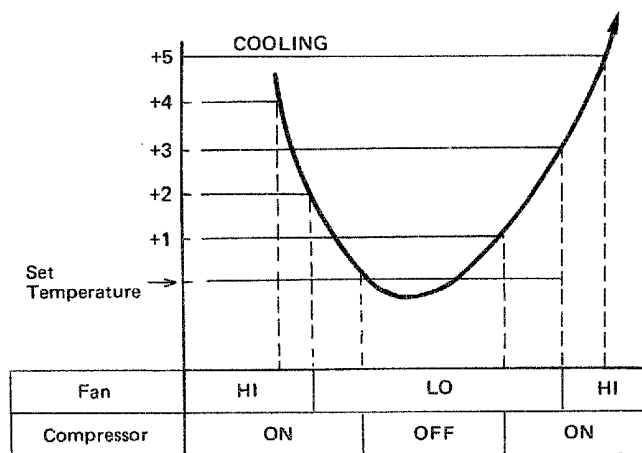
- 2) Frosting prevention (in cooling operation)
When the temperature of indoor unit heat exchanger drops below -5.5°C due to decreased load in the room in cooling operation, the compressor is stopped to prevent frosting of the heat exchanger.

7-2. Automatic Air Flow Control (indoor unit fan)

1) Auto fan select

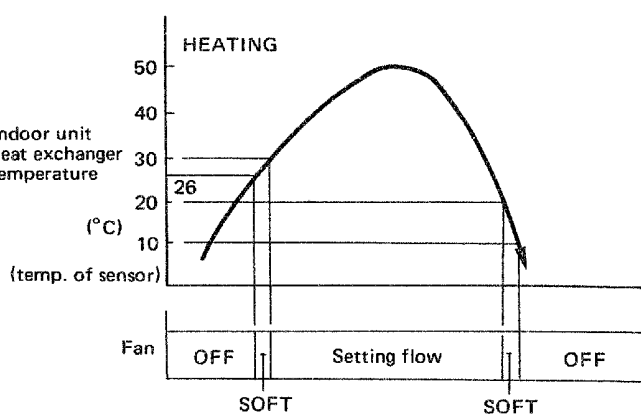
When SELECTOR switch is set to AUTO, the air flow is automatically controlled in 2 steps, HI (high) and LO (low), according to the difference between the THERMO setting point and the room temperature.

When the compressor stops because of thermostat operation in heating operation, the air flow is automatically changed to SOFT irrespective of the SELECTOR setting so as to prevent cold draft.



3) Cold Draft Prevention—Soft start heating (in heating operation)

At heating start-up or restarting after defrosting, the fan is stopped until the temperature of indoor unit heat exchanger exceeds 26°C to prevent cold draft. The fan starts at the SOFT flow rate when the temperature of heat exchanger exceeds 26°C and shift to the setting position of SELECTOR, when it exceeds 30°C.



4) Defrost-Indoor fan stop (in defrost operation)

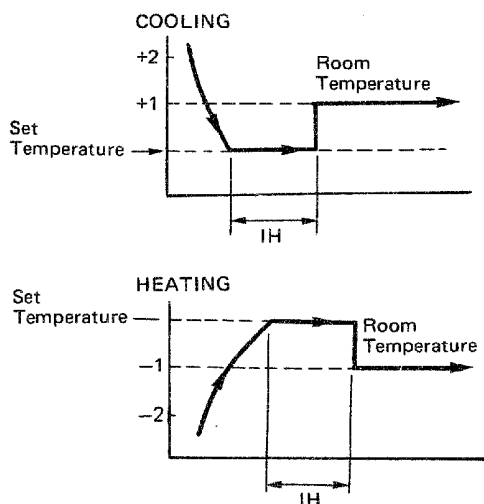
When defrost operation starts, blowing continues for 10 seconds to cool off the residual heat of the heat exchanger, then the fan stops. (This feature prevents waste energy and cold draft)

RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000HE8
RAV-1250CHE8/S1250HE8

7-3. Energy Saving Circuit (moderate power saver)

When the THERMO switch is set to SAVE, the micro-computer begins to keep the room temperature at 1 deg. (°C) above the set point while cooling or at 1 deg. (°C) below the set point while heating, about 1 hour after the room temperature has reached THERMO set temperature. This circuit prevents wasteful overcooling or overheating caused by too low setting while cooling, or to high setting while heating when initial start-up.

It saves 10–16% of cooling energy (at the ambient temp. of 32°C, to room temp. 27→28°C) or about 5% of heating energy (at the ambient temp. of 0°C, to room temp. 20→19°C)



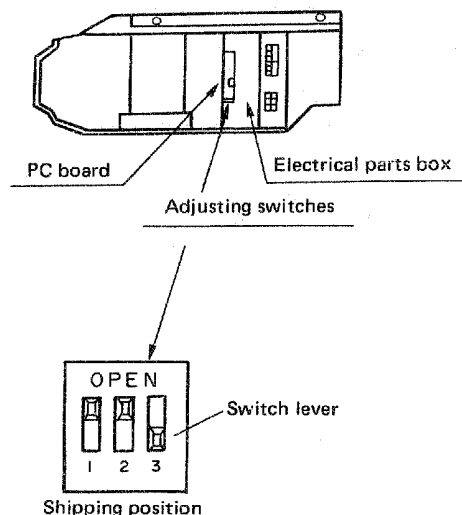
7-4. Temperature Shift (recalibration)

Expanded heating temperature range

Although the upper limit of the room temperature in heating operation is 28°C, this setting range may be insufficient depending on the room structures. The temperature setting range can be changed to enable use at maximum 35°C.

To change the temperature setting range

The adjusting switches are located on the PC board in the electric parts box of indoor unit. Change the temperature setting range according to the room conditions. Green colored 3 adjusting switches are as shown above.

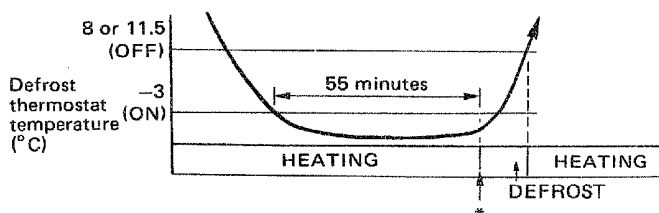


Switch position	Shift temp.
	2deg up
	4deg up
	6deg up

7-5. 4-way Valve Control

- 1) In heating operation: 4-way valve is always ON.
- 2) In defrost operation
4-way valve is OFF at 55 minutes after the defrost thermostat actuation. 4-way valve is ON upon end of defrost operation by thermostat resetting or lapse of 10 minutes. (maximum defrost time is 10 minutes)
- 3) When operation is stopped:
4-way is OFF at 3 minutes after the ON/OFF switch is pressed.

NOTE: The indoor fan keeps running for 10 seconds after the defrost operation starts.



*Defrost operation starts when the total defrost thermostat ON time reaches 55 minutes. [Efficient defrost cycle]

7-6. High Pressure Release, High Pressure Switch operation

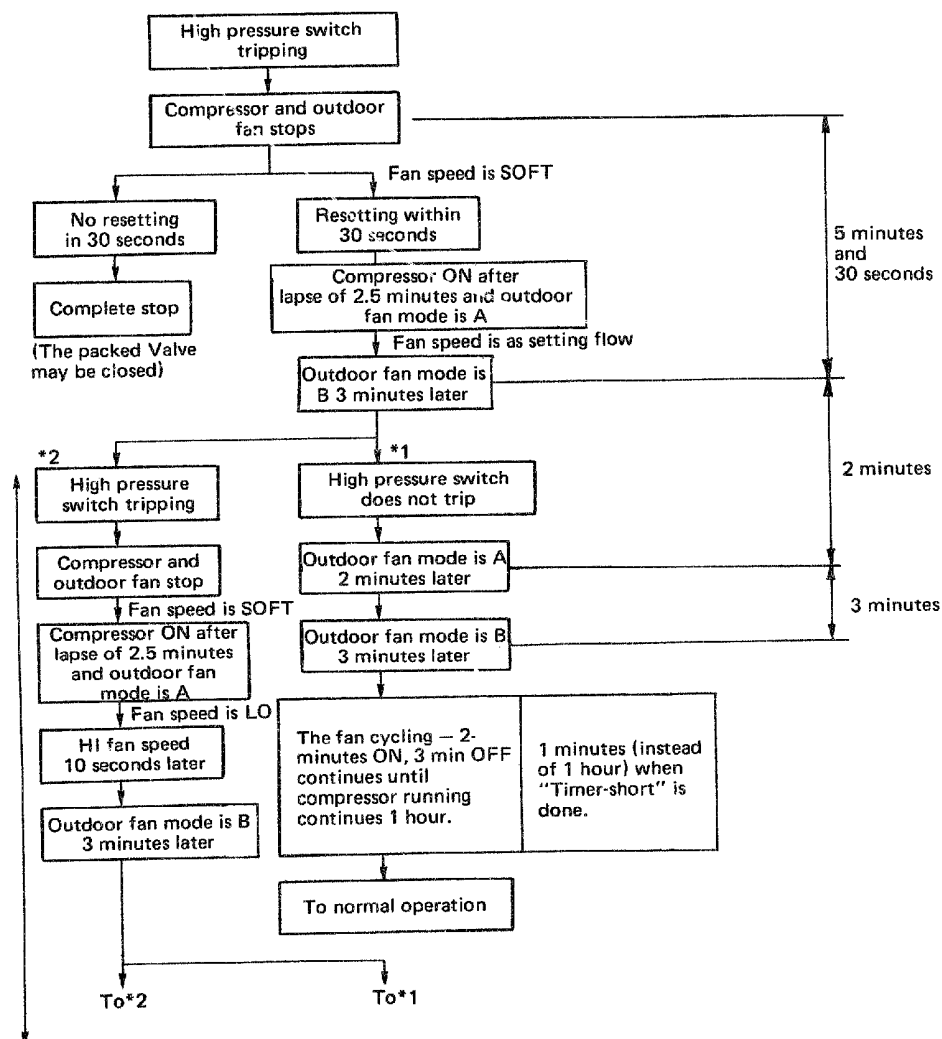
- 1) In cooling operation
Unit completely stops when the high pressure switch operate.
- 2) In heating operation
 - Outdoor fan release:
If the indoor heat exchanger temperature (TC) is 54°C or above, the outdoor fan operates mode A (as shown in the schedule in the next page) for 3

RAV-450CHJE/S450HJ
RAV-560CHJE/S560HJ
RAV-710CHE8/S710HE
RAV-1000CHE8/S1000
RAV-1250CHE8/S1250

Model		RAV-S450HJE RAV-S560HJE RAV-S710HE8	RAV-S1000HE8 RAV-S1250HE8	
Outdoor fan Operation mode	A	(FM ₀) stop	(FM ₀₁) (upper side): Shift to low flow (FM ₀₂) (lower side): Stop	
	B	(FM ₀) run	(FM ₀₁) (upper side): High flow run (FM ₀₂) (lower side): run	Normal

After 3 minutes, the outdoor fan will restart (as mode B) if TC is 50°C or below. (It will remain mode A if TC is still more than 50°C)

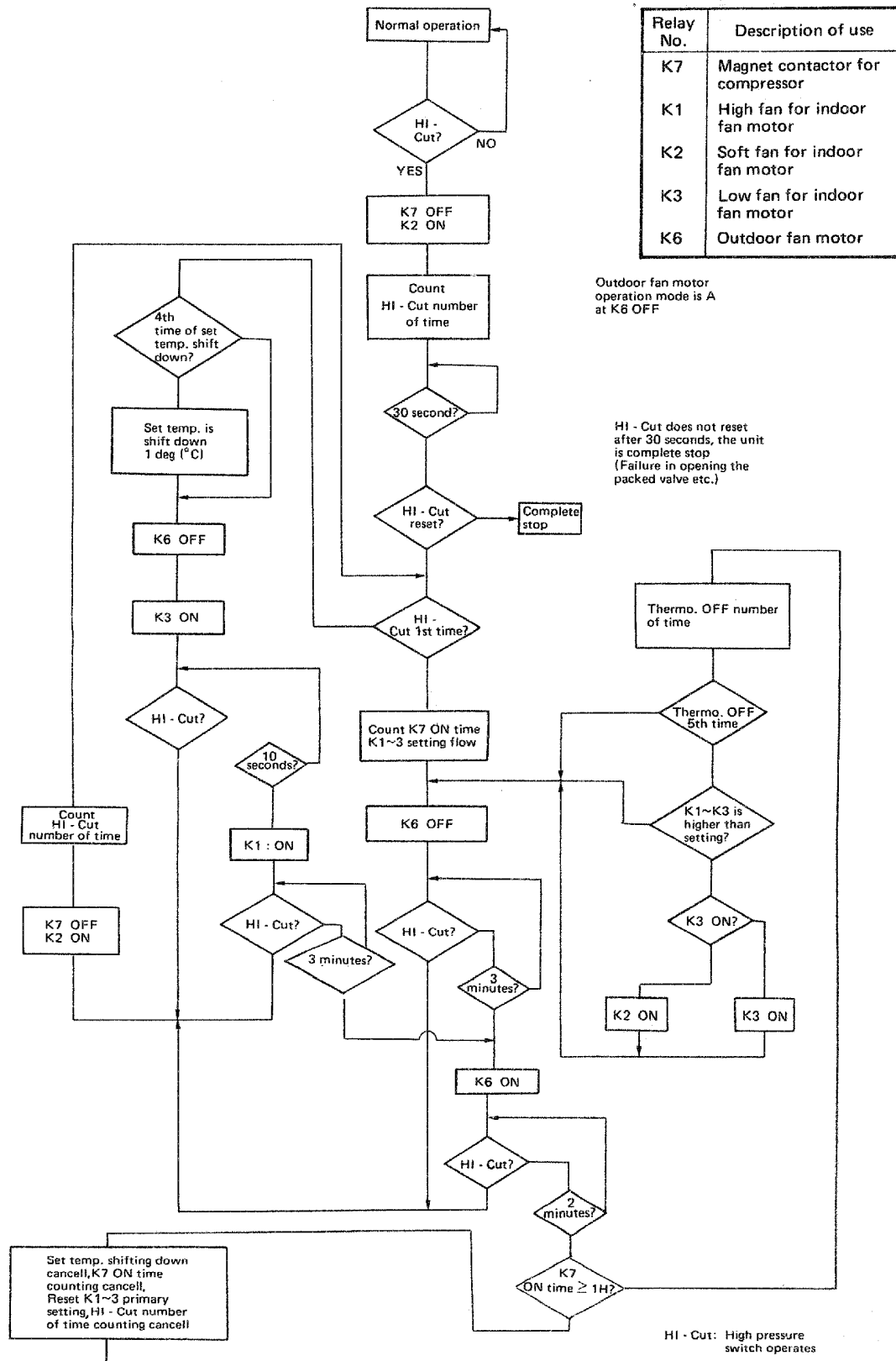
- **High pressure switch tripping:** see following schedule.



NOTE: The set temperature is shifted by 1°C up to 4°C each time the switch trips out. (High pressure switch operation count and release for indoor fan setting are omitted.)

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

High pressure switch operation flowchart



Relay No.	Description of use
K7	Magnet contactor for compressor
K1	High fan for indoor fan motor
K2	Soft fan for indoor fan motor
K3	Low fan for indoor fan motor
K6	Outdoor fan motor

RAV-450CHJE/S450
RAV-560CHJE/S560
RAV-710CHE8/S710
RAV-1000CHE8/S10
RAV-1250CHE8/S12

7-7. Filter Maintenance Sign

The filter lamp lights when cumulative time of the indoor fan operation has reached 120 hours.

When the filter RESET button (on the remote controller) is pressed, the filter lamp is cleared off.

Regardless of whether the lamp lights or not, the time cumulation is cancelled if the RESET button is pressed.

7-8. Timer-Short Indication

The filter lamp flickers for 12 minutes when the timer is shorted. (Changes to continuous lighting after lapse of 12 minutes).

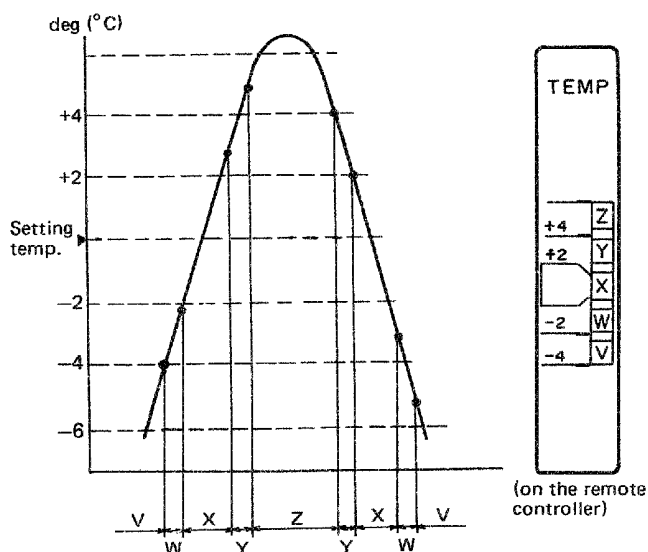
○ Indication Lamp Lighting Status

○: continuous lighting ●: flickering X: OFF

	Power OFF	Power ON	Operation	Defrost cycle cold draft prevention	Fan motor 12-hour operation	Timer-short	Residual heat removal (Optional heater)
ON/OFF lamp	X	●	○	○	X	X	● (maximum 2 minutes)
FILTER lamp	X	X	X	X	○ (X Reset)	● (○ after 12 minutes)	X
DEFROST lamp	X	●	X	○	X	X	X

7-10. Present Room Temperature Monitor

TEMP lamp lights based on the difference between the set temperature and the present room temperature read by sensor for room temperature. (No lamp lights while timer for optional heater operates)



NOTE: Lamp x lights continuously

7-9. Power Failure Indication

The ON/OFF (operation) lamp and DEFROST lamp flashes at 1 second intervals when the reset status is provided at the time of main power switch is turned on or the power resume after interruption of service.

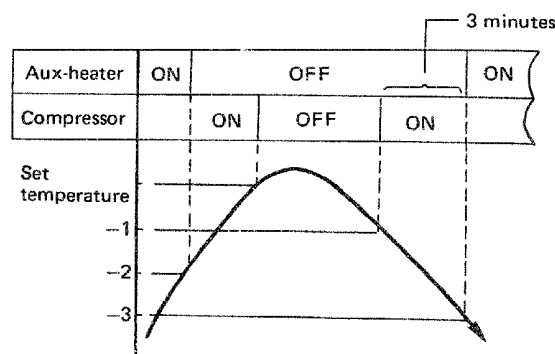
7-11. Optional Auto-Turn Louver Control

If the motor kit (option) is installed, the circuit controls relay for auto-turn louver when the unit is operated and auto-turn louver switch is turned ON.

7-12. Optional Aux-Heater Control (Independent control)

When the temperature approaches the THERMO setting, the aux-heater is OFF firstly and then compressor is OFF. This 2-step control minimized use of the aux-heater for full attainment of the heat pump economy.

To prevent recycle of the aux-heater, power is not supplied to aux-heater for 3 minutes after the compressor is re-started.



RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000HE8
RAV-1250CHE8/S1250HE8

(In case of using aux-heater it is necessary to turn ON
aux-heater switch on the PC board.

The aux-heater power is supplied at 10 seconds after the
compressor starting to decrease the rush current.

7-13. Optional Aux-Heater Residual Heat Removal

Effective in heating operation only.

When the unit is stopped with the aux-heater power at ON,
the indoor fan is operated at SOFT flow rate for 2 minutes
after the aux-heater power OFF and the ON/OFF (opera-
tion) lamp flashes at 1 second intervals.

(Example) The fan stops 2 minutes later after operation
is stopped when the aux-heater is powered.
The fan stops 1 minute later when operation
is stopped at 1 minute after the aux-heater
is turned OFF.

RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE!
RAV-1000CHE8/S1000H
RAV-1250CHE8/S1250H

8. OPERATION

8-1. Operation and Time

	Operation	Operation time	Fan	Cooling operation	Heating operation	Operation time with Timer-short
1	Main power switch ON	Zero second 1 minute 10 seconds	ON/OFF lamp and DEFROST lamp flicker			FILTER lamp flickers
2	ON/OFF button ON (press)		ON/OFF lamp lights, DEFROST lamp OFF			
			Indoor fan ON at LO by K3 operation	Bar display for room temp. monitor ON		
				Indoor fan ON by operation of K1, K2 and K3 based on the set temperature	4-way valve ON by K9 operation Indoor fan ON at SOFT by K2 operation	
				Compressor ON by K7 operation Outdoor fan ON by K6 operation		
					The indoor fan operates according to the indoor heat exchanger temperature. See page 19. The DEFROST lamp lights in the fan OFF zone. Indoor fan is operated based on the set temperature.	
		When the heater switch (on the PC board) is ON, optional aux-heater ON by K11 operation. However, it is not turned ON when the indoor fan is set OFF.				
	AUTO-TURN LOUVER switch ON		Optional geared motor for auto-turn louver ON by K13 operation only when the indoor fan ON.			
	SELECTOR switch at AUTO or LO or HI			Operation starts according to the switch position.		
	THERMO switch, at SAVE	1 hour		When the compressor stops due to the room temperature reaching the set temperature, the unit is operated for 1 hour at the set temperature and then set back 1 deg. higher in cooling or 1 deg. lower in heating.		1 minute
	RESET button (press)	120 hours	When the total operation time of the indoor fan motor reaches 120 hours, the FILTER lamp lights. (Turning OFF the main power switch before 120 hours will set back to zero) Clean the filter and press the RESET button to turn off the FILTER lamp			12 minutes
	High pressure switch operation		Complete stop	Explained on page 20.		

Cont'd.

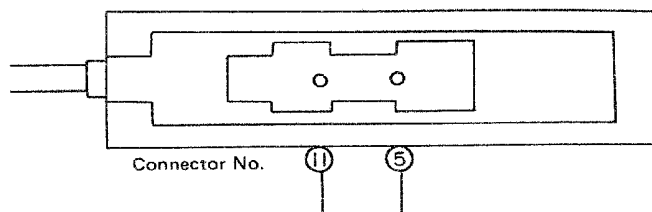
RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

	Operation	Operation time	Fan	Cooling operation	Heating operation	Operation time with Timer-short
	Defrost cycle	55 minutes 10 minutes 10 second			When the defrost thermostat operates, counting of the operation time starts. When 55 minutes passed, K9 and K6 are operated to turn OFF the 4-way valve and outdoor fan. The Indoor fan motor is turned OFF 10 seconds later. When the outdoor heat exchanger temperature rises to reset the defrost thermostat, K9 and K6 are actuated to operate the 4-way valve and outdoor fan. The indoor fan operates according to the indoor temperature. Defrost cycle does not end in 10 minutes after the start, it is terminated and heating operation restarts.	9 minutes 10 seconds 10 seconds 1 minute 40 seconds
3	ON/OFF button OFF (press again)	Zero second 2 minutes 3 minutes	Indoor fan stops by K3 operation, ON/OFF lamp OFF	Compressor, indoor and outdoor fan stop by K1 ~ K7 operation, ON/OFF lamp OFF	If the optional aux-heater is installed, K2 is actuated to operate the indoor fan at SOFT for 2 minutes. The ON/OFF lamp flickers during this period. (This operation does not occur if the aux-heater has been turned OFF by thermostat.) K9 is reset 3 minutes later to turn OFF the 4-way valve	2 minutes 3 minutes
4	Press the RUN/STOP switch again to operation after stop	2 minutes 30 seconds	Return to 1	The compressor and outdoor fan are turned ON when 2½ minutes have passed after they were stopped.		Zero second

8-2. Timer-short

Each operation time can be shortened as show in 8-1 to reduce the time for microcomputer function check. Since the compressor safety start timer is also shortened when timer-short does, remove the excitation of the magnetic contactor for the compressor before doing timer-short.

Connect a diode (1S1555 or equivalent) to the holes shown in the figure below on the rear side of the remote control switch.



The filter lamp flickers for 12 minutes when the Timer-short is done. (Changes to continuous lighting after lapse of 12 minutes.)

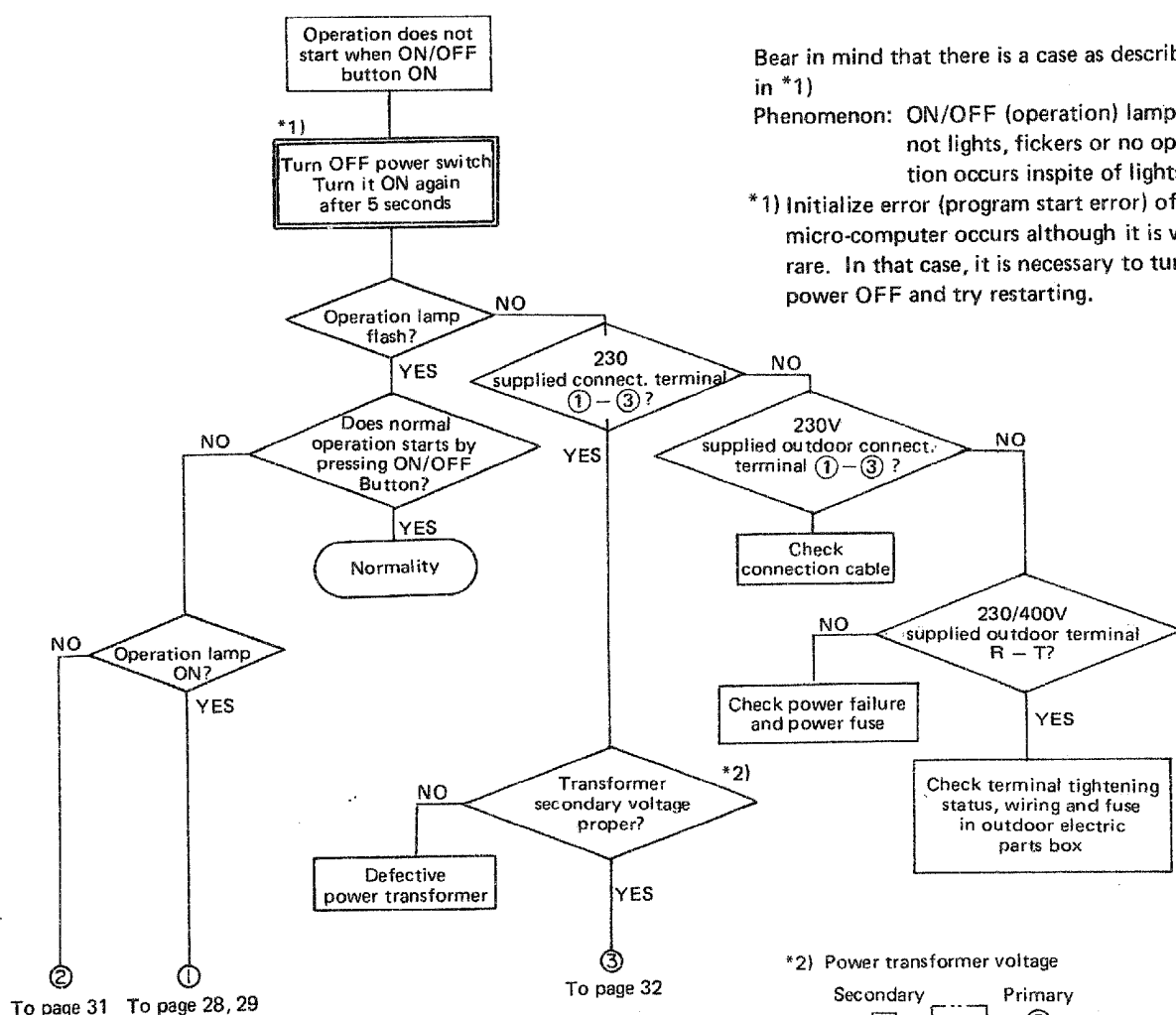
9. TROUBLESHOOTING CHART

9-1. Procedure for Microcomputer diagnosis

Since electronic control (microcomputer control) uses many parts and small current signals, it is likely to be affected by external noise as compared with the conventional circuit in which the power supply circuit is directly controlled by mechanical switches.

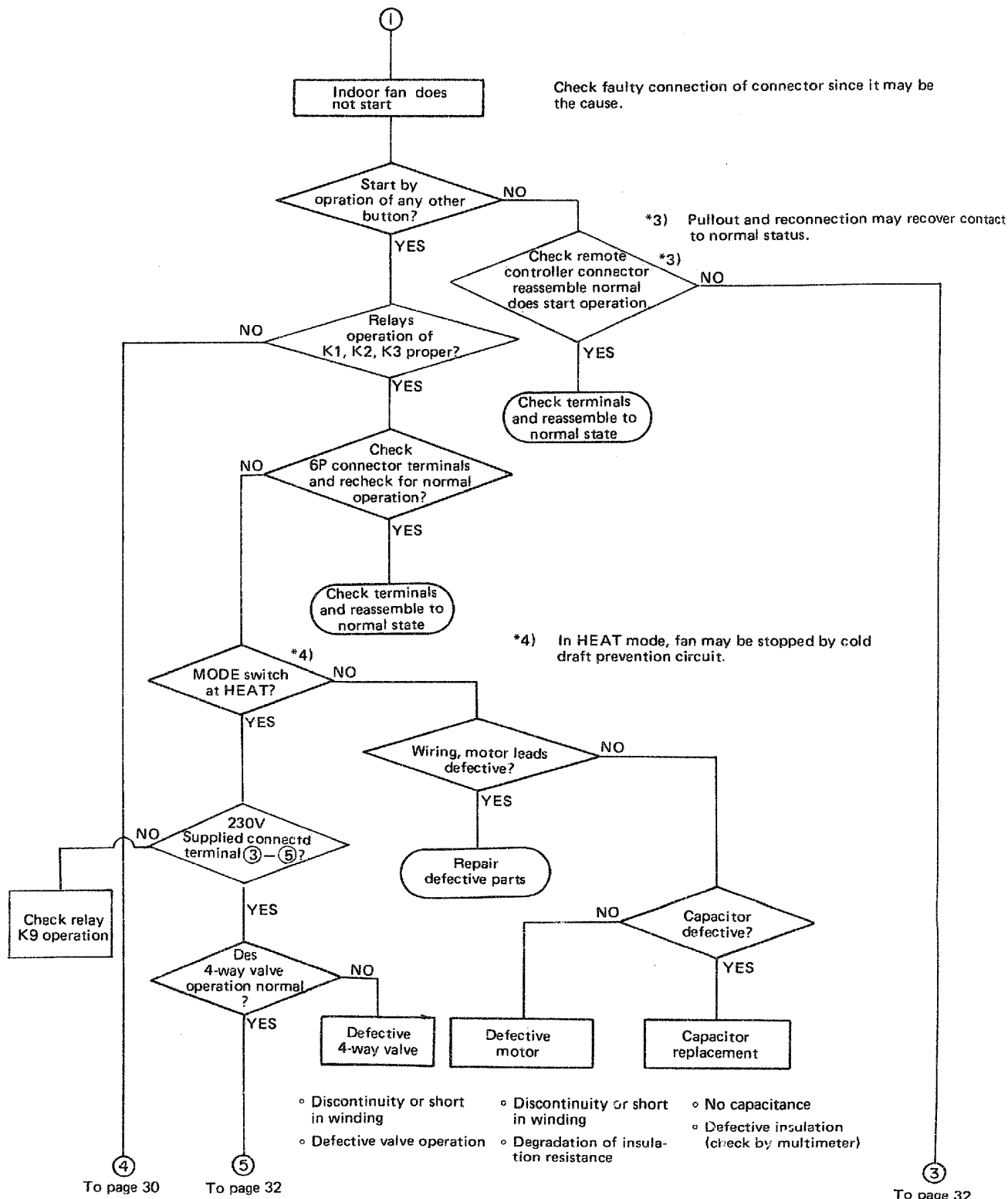
- Troubles with microcomputer itself (Reset error and defective parts).
- Defect of connecting parts (Noise caused by faulty connector connection, loose main power terminals, and loose connections of mini-connectors for remote controller switch, sensors and transformers).

(1) Operation does not start when ON/OFF controller is pressed

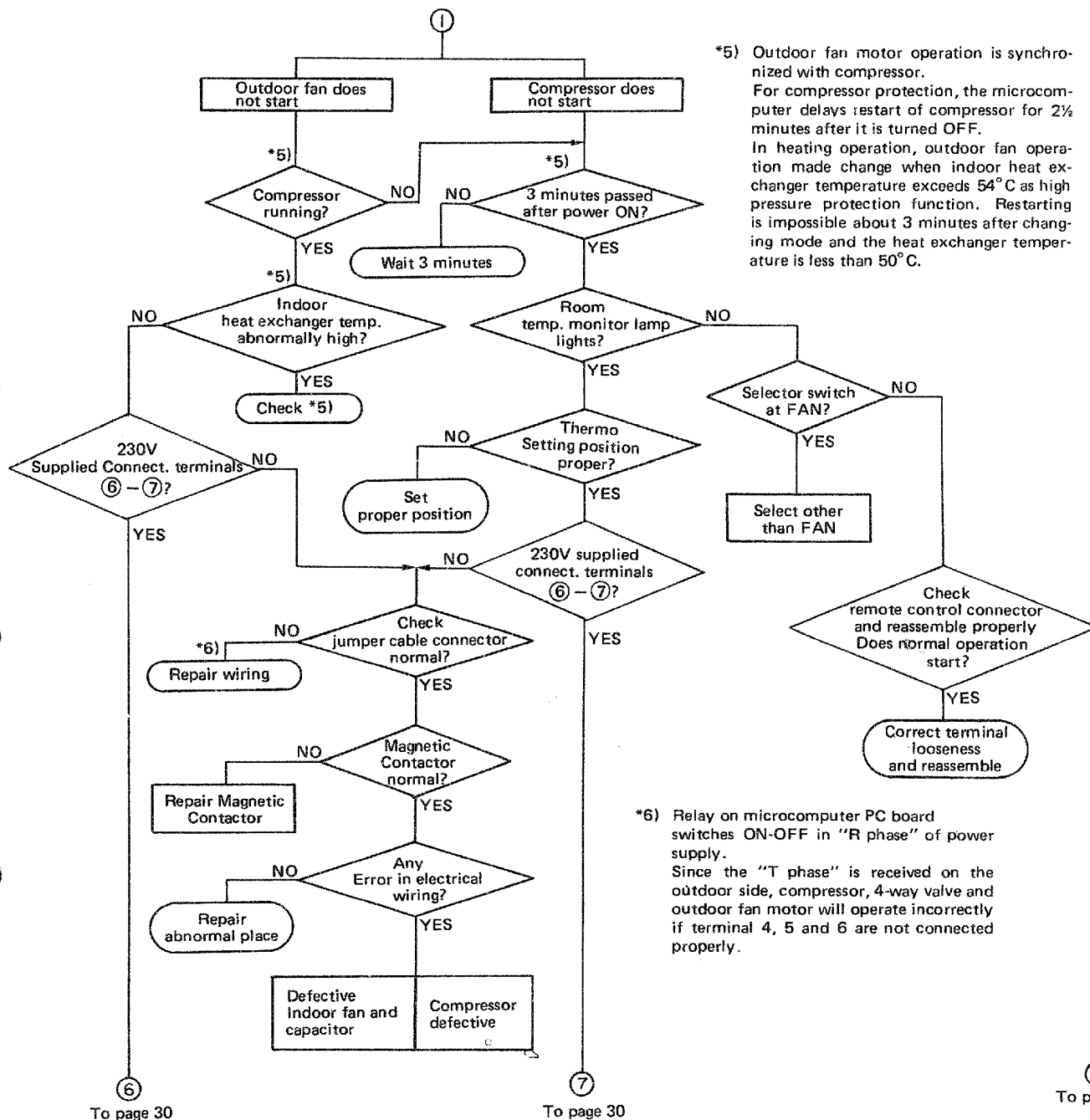


RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000HE8
RAV-1250CHE8/S1250HE8

(2) Indoor fan does not start . . . Check on indoor fan motor circuit

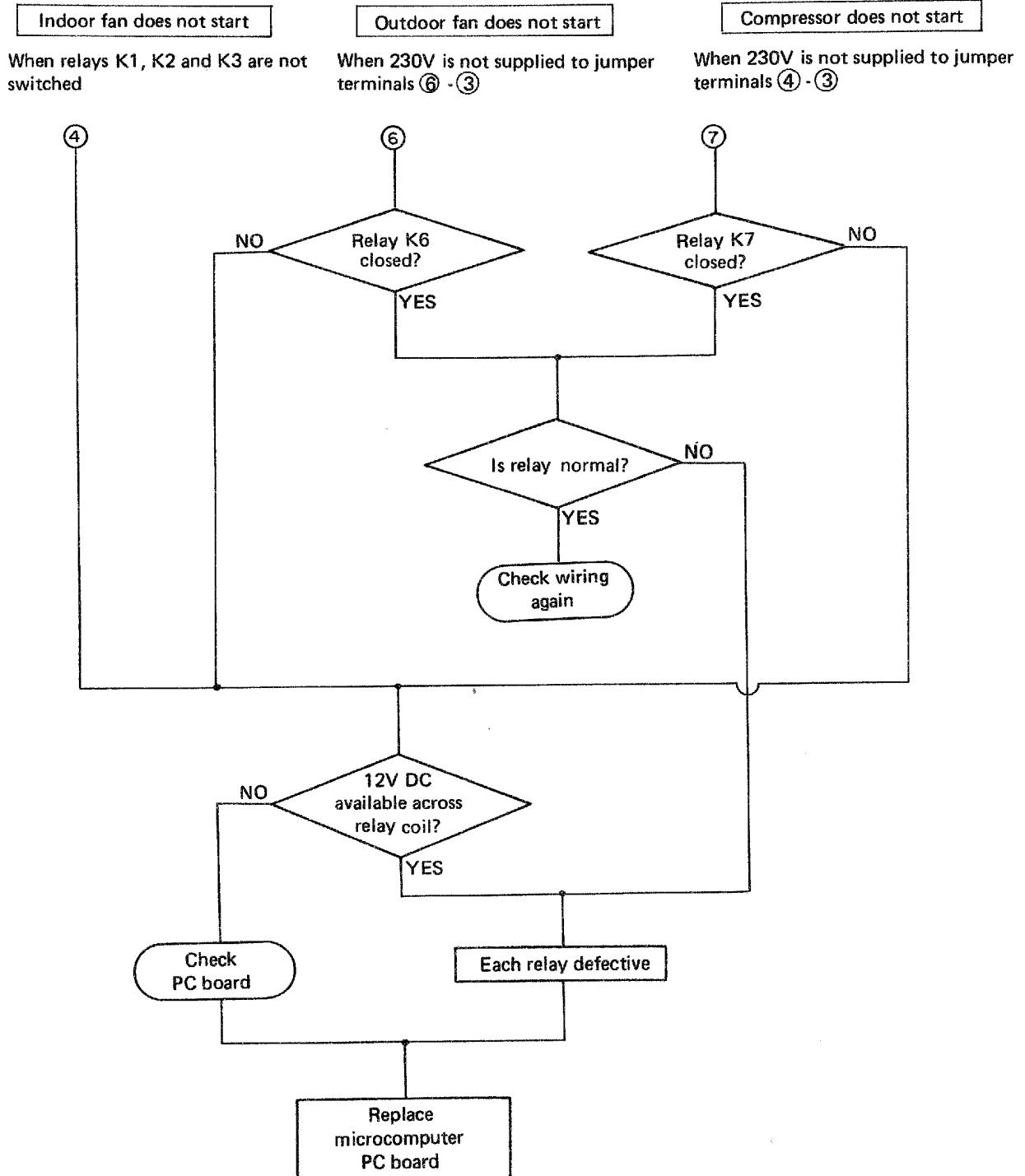


(3) Cooling (Heating) operation does not start . . . Check on outdoor fan and compressor circuit



RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

(4) Insufficient cooling or heating . . . Checking relays of indoor and outdoor fan motors and compressor

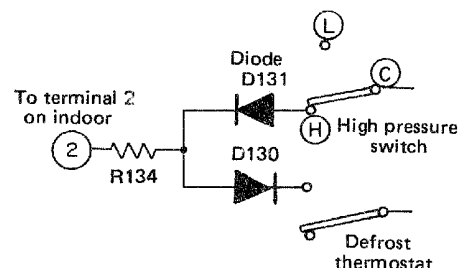
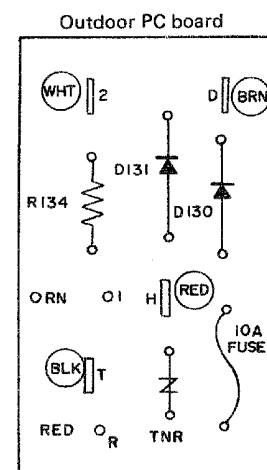
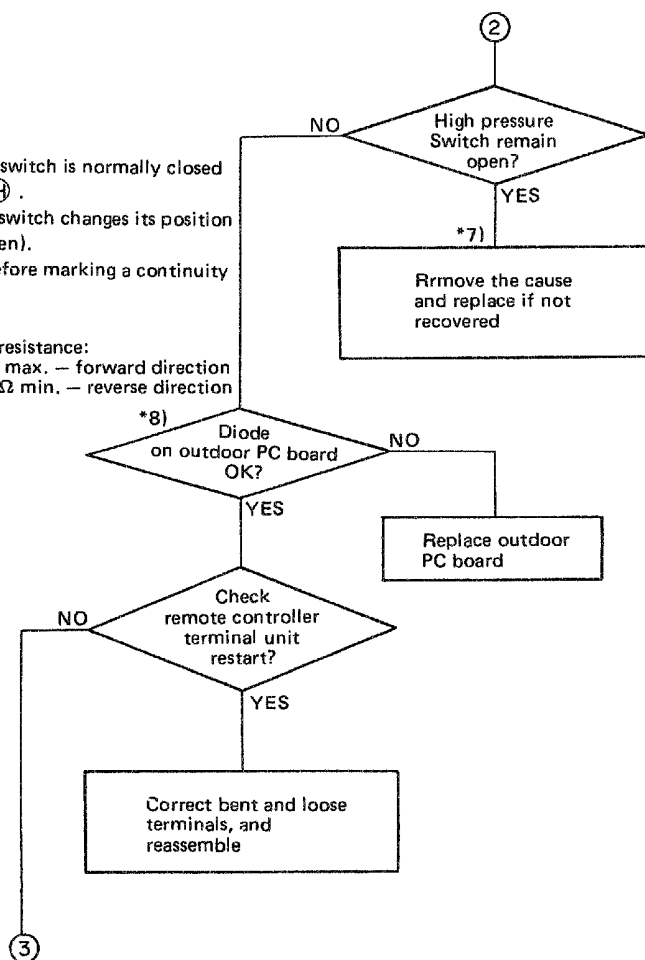


RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000HE
RAV-1250CHE8/S1250HE

(5) Operation does not start . . . Checking high pressure switch and outdoor PC board

*7) The high pressure switch is normally closed between C - H .
When it trips, the switch changes its position to C - L (open).
Turn power off before marking a continuity test.

Diode resistance:
200Ω max. — forward direction
100KΩ min. — reverse direction



*7) *8) When high pressure switch is open, no current flows to D131. As a result, a signal is sent to CPU from photocoupler connected to ② to inform high pressure switch cutout. When the diode is fused, operation mode becomes the same as when high pressure switch is open.

11-pin mini-connector is used to connect the remote controller switch to PC board.

Since each operation signal is sent from remote controller, imperfect connections between remote controller and PC board will cause malfunction.

Most of the allegedly defective microcomputer are often found not really faulty when checked at factory. Before returning microcomputer, be sure the contact is adequate.

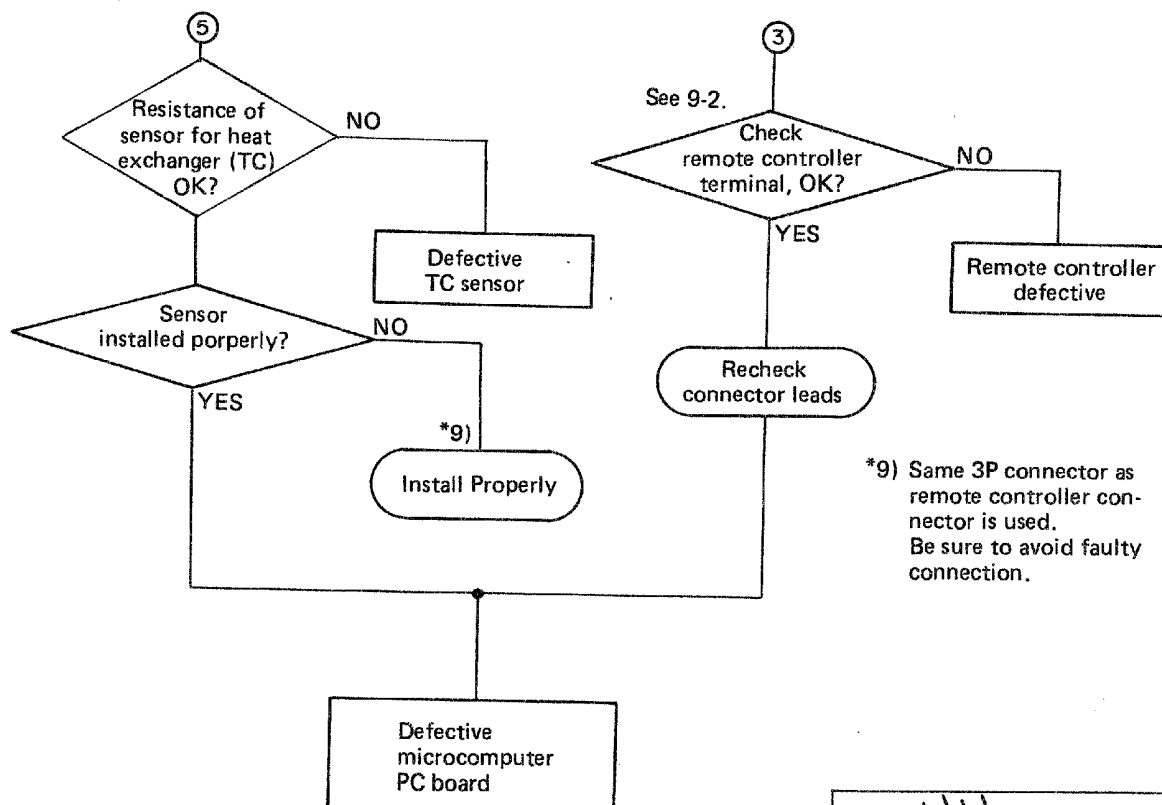
Although connector quality is sufficiently reliable, long repeated use may cause poor contact.

Pull out connector once and clear its surface to reduce contact resistance.

Malfunction may be eliminated. Coating contact restoring agent such as CRC will further improve effect.

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

(6) Operation does not start . . . Checking the remote controller and PC board

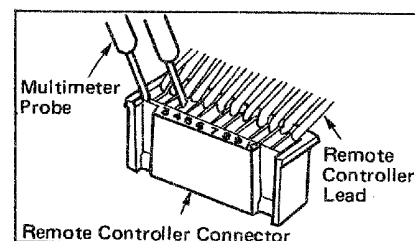


9-2. Remote Controller check method

Remote Controller function can easily be checked as follows:

Remove remote controller connector from PC Board and check continuity between terminals of remote controller connector by multi-meter.

(Procedure)

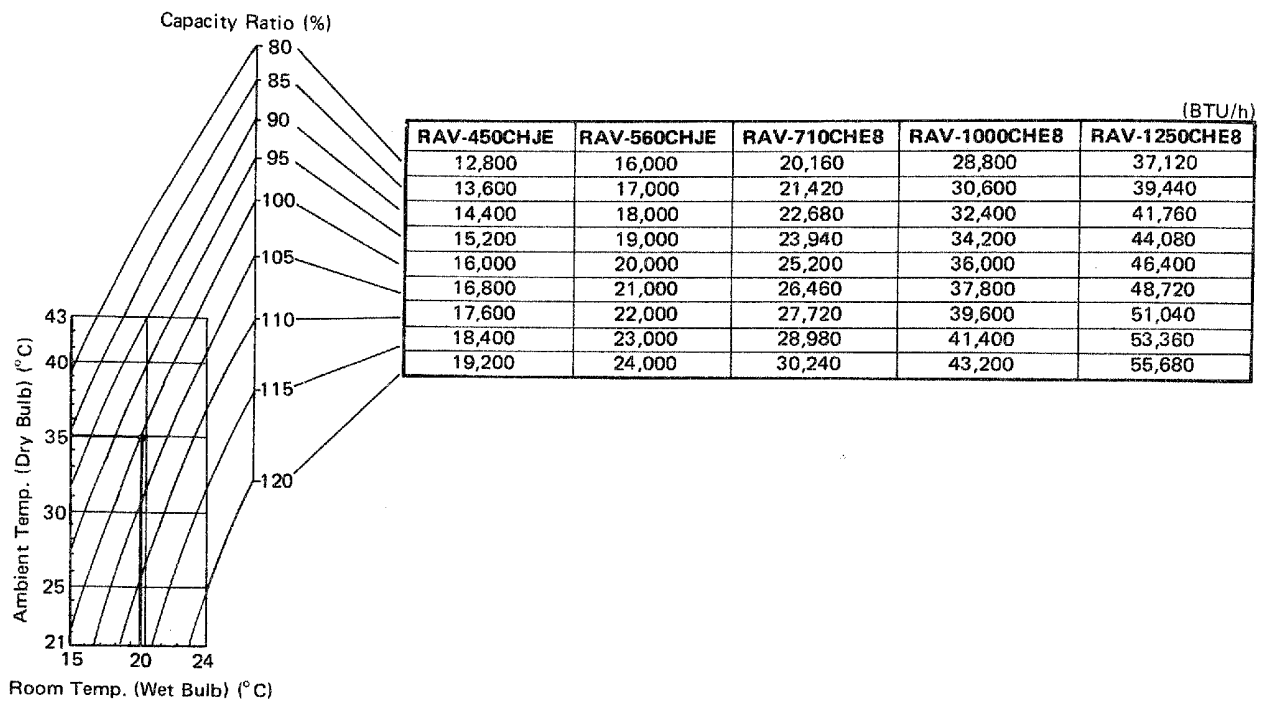


No.	Multimeter probe		Remote controller switch key position	Criterion
	Black ⊖	Red ⊕		
1	⑧ Gray	⑦ Brown ④ Orange	THERMO setting switch: COOL 25 ~ 30 ON/OFF switch: Kept pressed	Continuity shall exist Several hundred ohms
2	⑨ Pink	⑦ Brown ⑥ Black ⑤ White ④ Orange	THERMO. setting switch: COOL 23 ~ 26 Fan speed SELECTOR switch: LO THERMO. set-back switch: SAVE FILTER RESET Button: Kept pressed	
3	⑩ Blue	⑦ Brown ⑥ Black ⑤ White ④ Orange	THERMO. setting switch: COOL 22, 23, 26, 27, 30, 31 Fan speed SELECTOR switch: HI AUTO-TURN LOUVER switch: ON MODE-switch: HEAT	
		⑦ Brown	THERMO. setting switch: COOL 29 ~ 32	

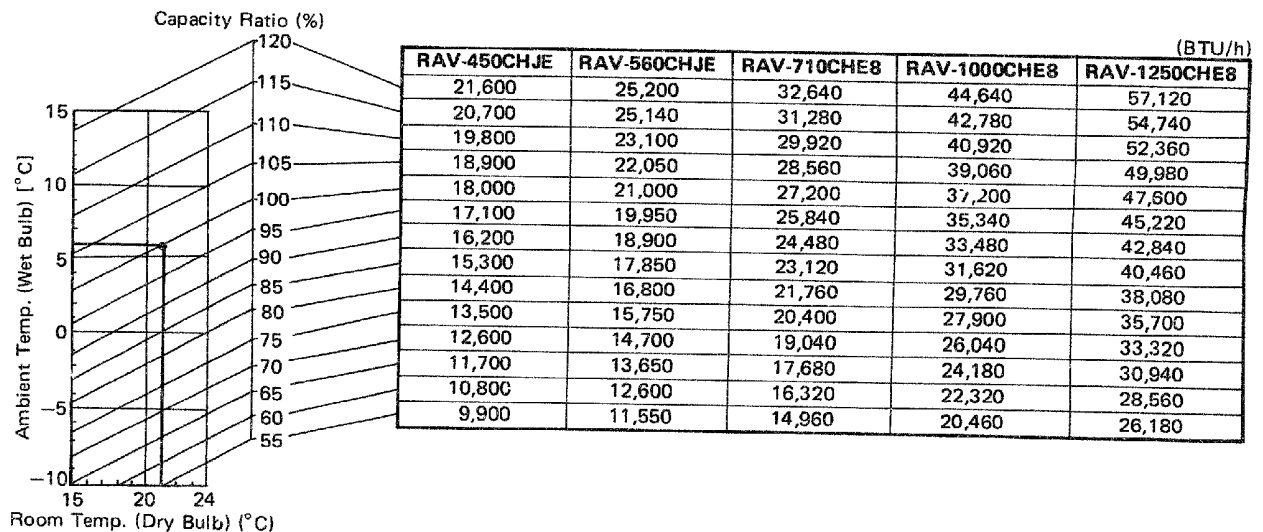
RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

10. PERFORMANCE CHARACTER

10-1. Cooling Capacity



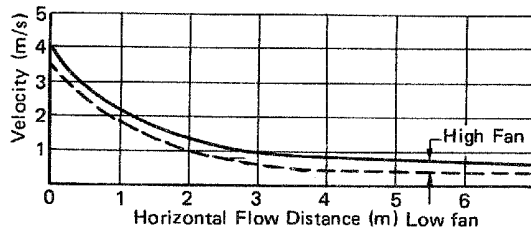
10-2. Heating Capacity



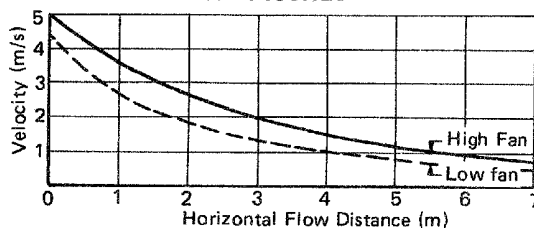
RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE
 RAV-1250CHE8/S1250HE

10-3. Supply Air Velocity and its Covering Distance

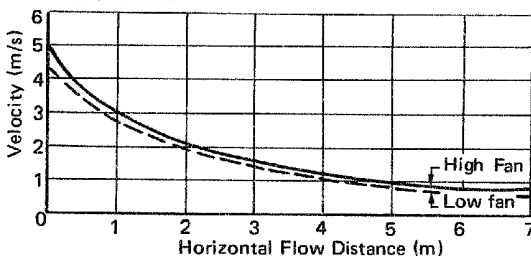
RAV-450CHJE



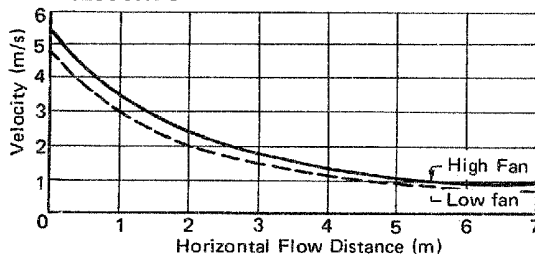
RAV-560CHJE RAV-710CHE8



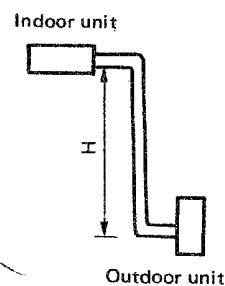
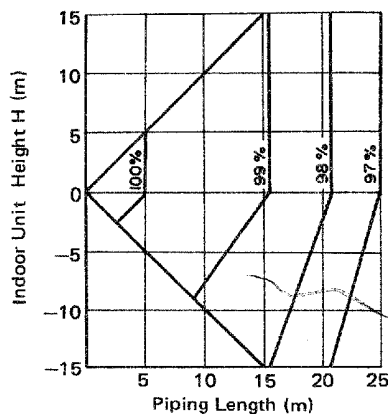
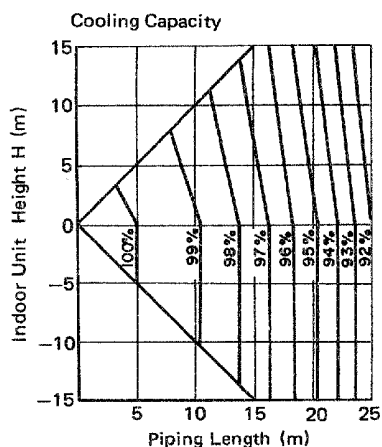
RAV-1000CHE8



RAV-1250CHE8



10-4. Piping Length/Cooling Capacity/Heating Capacity



10-5. Piping Length/Additional Refrigerant Volume

In case of piping length is more than 5m, less than 25m.

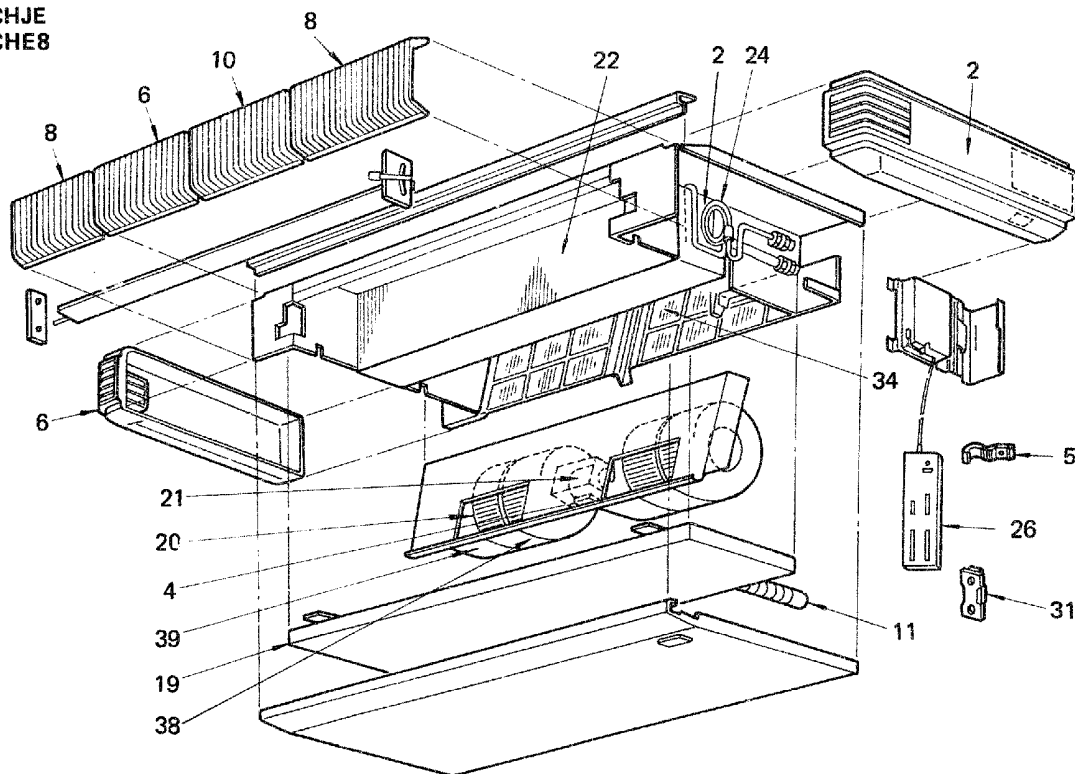
Model	Add. volume (per 1 meter)
RAV-450CHJE/S450HJE	30 g/m
RAV-560HJE/S560HJE	30 g/m
RAV-710CHE8/S710HE8	40 g/m
RAV-100CHE8/S1000HE8	50 g/m
RAV-1250HE8/S1250HE8	50 g/m

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

11. EXPLODED VIEWS AND PARTS LISTS

11-1. Indoor unit

RAV-450CHJE
 RAV-560CHJE
 RAV-710CHE8

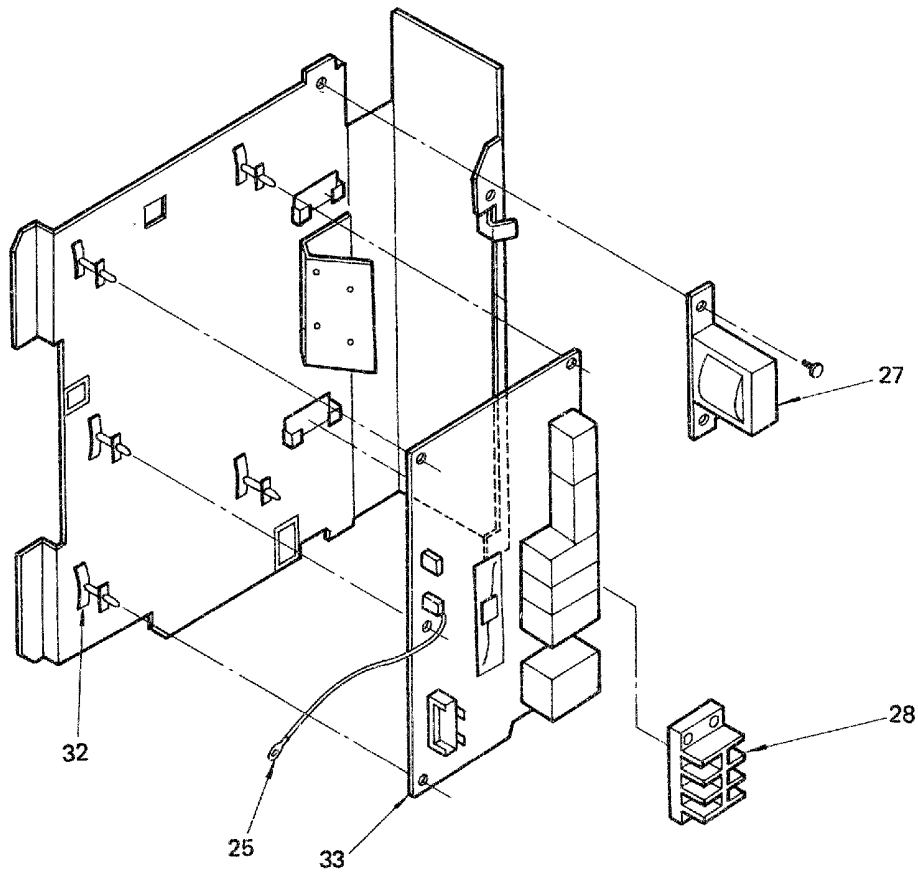


Location No.	Part No.	Description
2	43019604	Holder, Sensor
4	43055275	Capacitor, Electrolytic, 3MFD, 400V AC
5	43063135	Holder, Cord
6	43102505	Side, Panel
7	43102540	Side, Panel (RAV-450CHJE, 560CHJE)
7	43102541	Side, Panel CRAV-710CHE8)
8	43109200	Grille, Outlet
9	43109202	Grille, Outlet
10	43109201	Grille, Outlet (RAV-560CHJE, 710CHE8)
11	43199028	Hose, Drain
19	43191119	Drain Pan (RAV-450CHJE)
19	43191120	Drain Pan (RAV-560CHJE, 710CHE8)
20	43120074	Fan, Sirrocco (RAV-450CHJE)
20	43120095	Fan, Sirrocco (RAV-560CHJE, 710CHE8)
21	43121279	Motor, AC230V, 50Hz Fan (RAV-450CHJE)

Location No.	Part No.	Description
21	43121282	Motor, AC230V, 50Hz Fan (RAV-560CHJE, 710CHE8)
22	43144327	Evaporator (RAV-450CHJE)
22	43144376	Evaporator (RAV-560CHJE)
22	43144366	Evaporator (RAV-710CHE8)
24	43047492	Tube, Capillary (RAV-450CHJE)
24	43047491	Tube, Capillary (RAV-560CHJE, 710CHE8)
26	43169280	Remote Control, Assembly
31	43163014	Holder
34	43180212	Air Filter (RAV-450CHJE)
34	43180188	Air Filter (RAV-560CHJE, 710CHE8)
38	43723019	Fan Case, Right (RAV-450CHJE)
38	43126105	Fan Case, Right (RAV-560CHJE, 710CHE8)
39	43723020	Fan Case, Left (RAV-450CHJE)
39	43126104	Fan Case, Left (RAV-560CHJE, 710CHE8)

RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000HE8
RAV-1250CHE8/S1250HE8

RAV-450CHJE
RAV-560CHJE
RAV-710CHE8

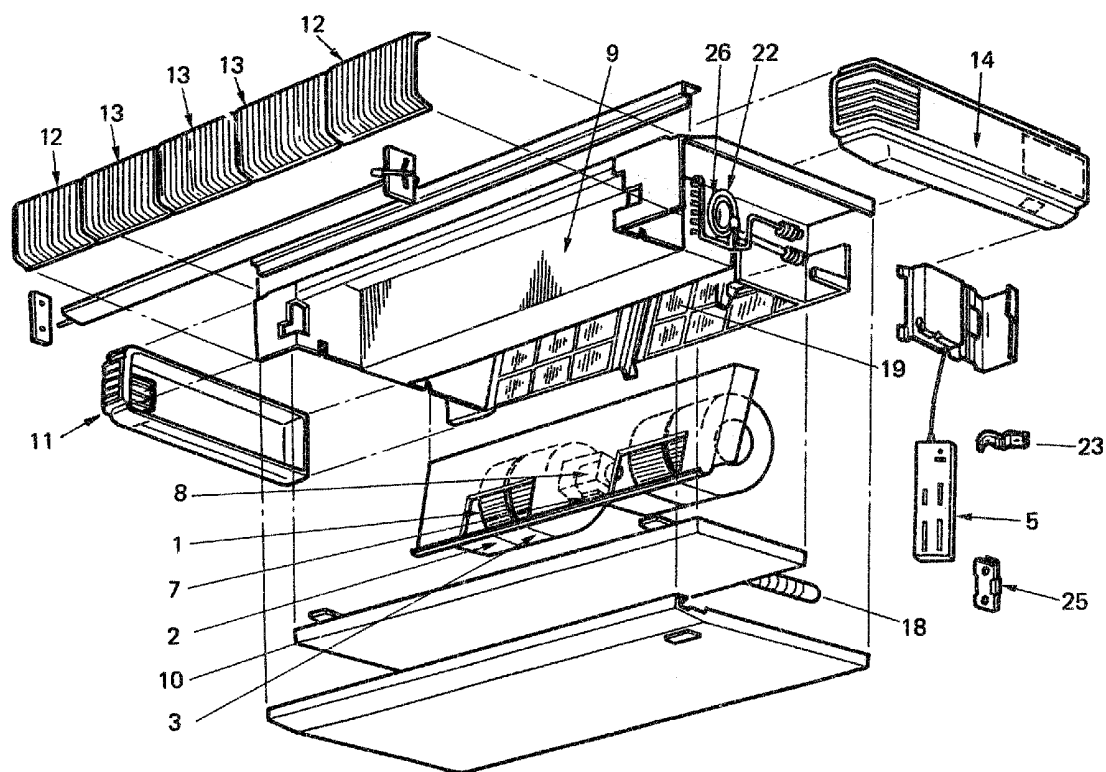


Location No.	Part No.	Description
25	43150104	Sensor, Room Temp.
27	43158066	Transformer, Power
28	43160261	Terminal Block

Location No.	Part No.	Description
32	43163015	Support
33	43169282	PC Board Assembly

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

RAN-1000CHE8
 RAV-1250CHE8

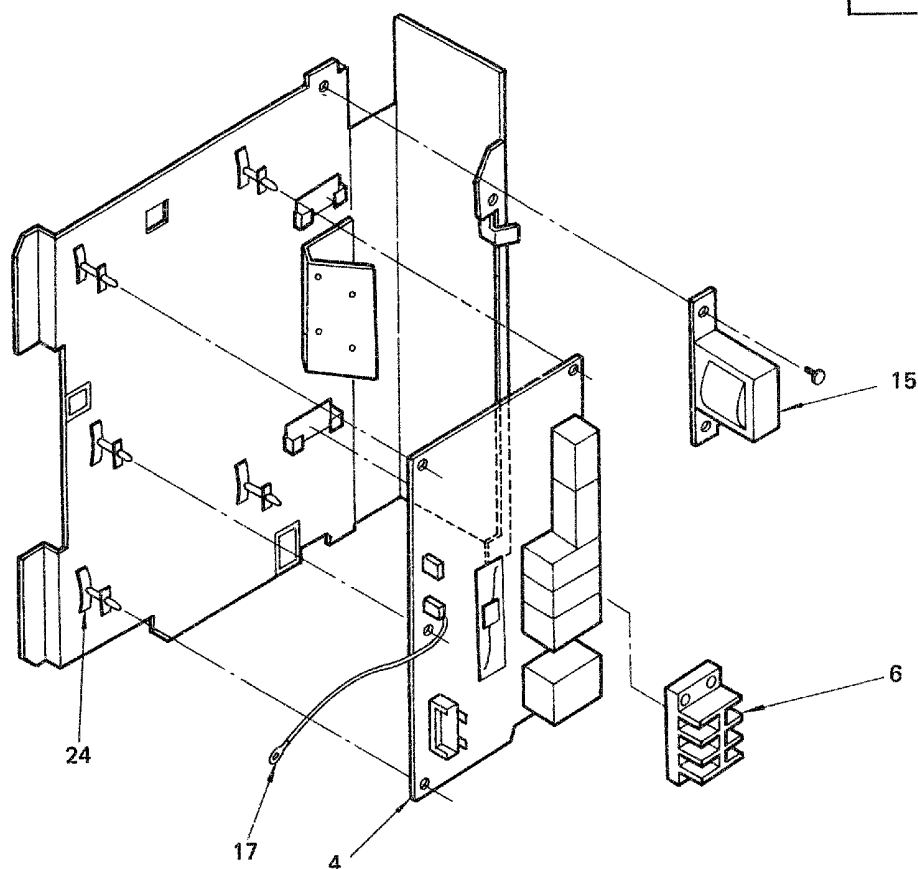


Location No.	Part No.	Description
1	43120095	Fan, Sirrocco
2	43126104	Fan Case, Left
3	43126105	Fan Case, Right
5	43169280	Remote Control, Assembly
7	43055274	Capacitor, Electrolytic, 4MFD, 400VAC (RAV-1000CHE8)
7	43155056	Capacitor, Electrolytic, 5MFD, 400VAC (RAV-1250CHE8)
8	43121283	Motor, 230V, 50Hz Fan (RAV-1000CHE8)
8	43121281	Motor, 230V, 50Hz Fan (RAV-1250CHE8)
9	43144413	Evaporator (RAV-1000CHE8)
9	43144414	Evaporator (RAV-1250CHE8)

Location No.	Part No.	Description
10	43191135	Drain Pan (RAV-1000CHE8)
10	43191136	Drain Pan (RAV-1250CHE8)
11	43102505	Side Panel, Left
12	43109200	Grille, Outlet
13	43109201	Grille, Outlet
14	43102544	Side Panel, Right
18	43199028	Hose, Drain
19	43180191	Air, Filter (RAV-1000CHE8)
19	43180188	Air, Filter (RAV-1250CHE8)
22	43047491	Tube, Capillary
23	43063135	Holder, Cord
25	43163014	Holder
26	43019604	Holder, Sensor

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE
 RAV-1250CHE8/S1250HE

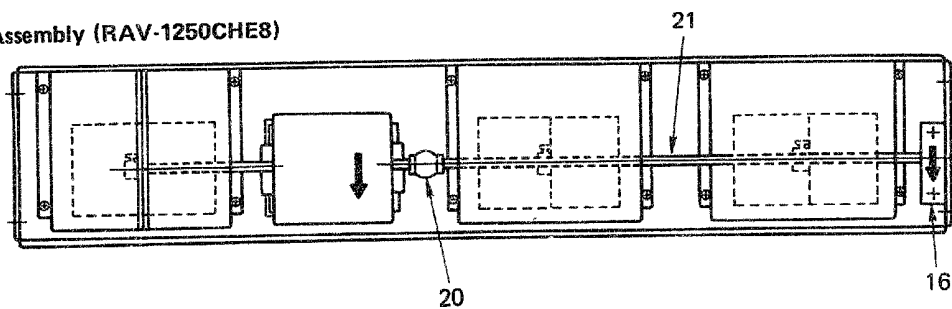
RAV-1000CHE8
 RAV-1250CHE8



Location No.	Part No.	Description
4	43169282	PC Board Assembly
6	43160261	Terminal Block
15	43158066	Transformer, Power

Location No.	Part No.	Description
17	43150104	Sensor, Room Temp.
24	43163015	Support

Blower Assembly (RAV-1250CHE8)



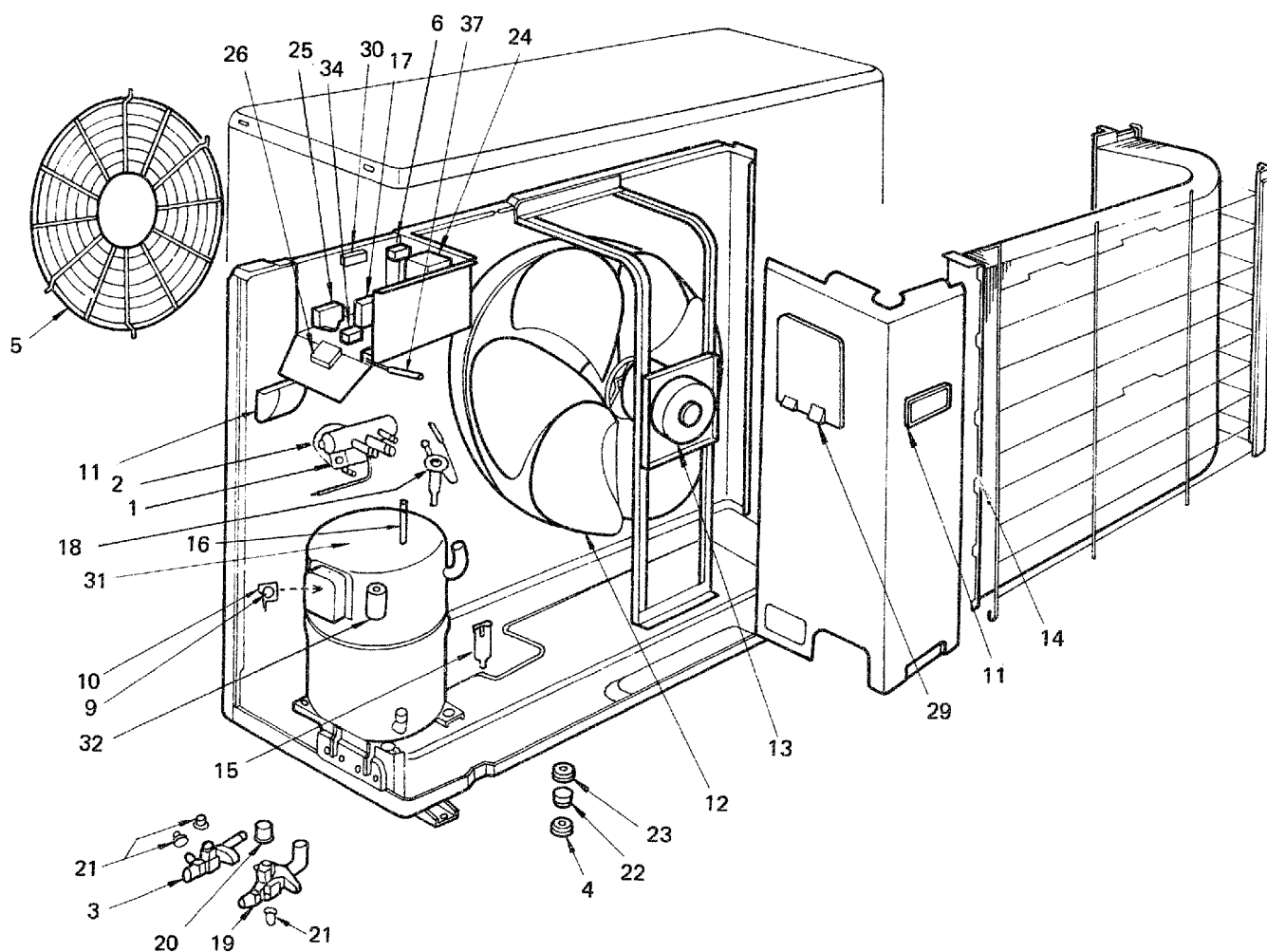
Location No.	Part No.	Description
16	43125123	Bearing
20	43125110	Coupling

Location No.	Part No.	Description
21	43125112	Shaft

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

11-2. Outdoor unit

RAV-S450HJE
 RAV-S560HJE
 RAV-S710HE8



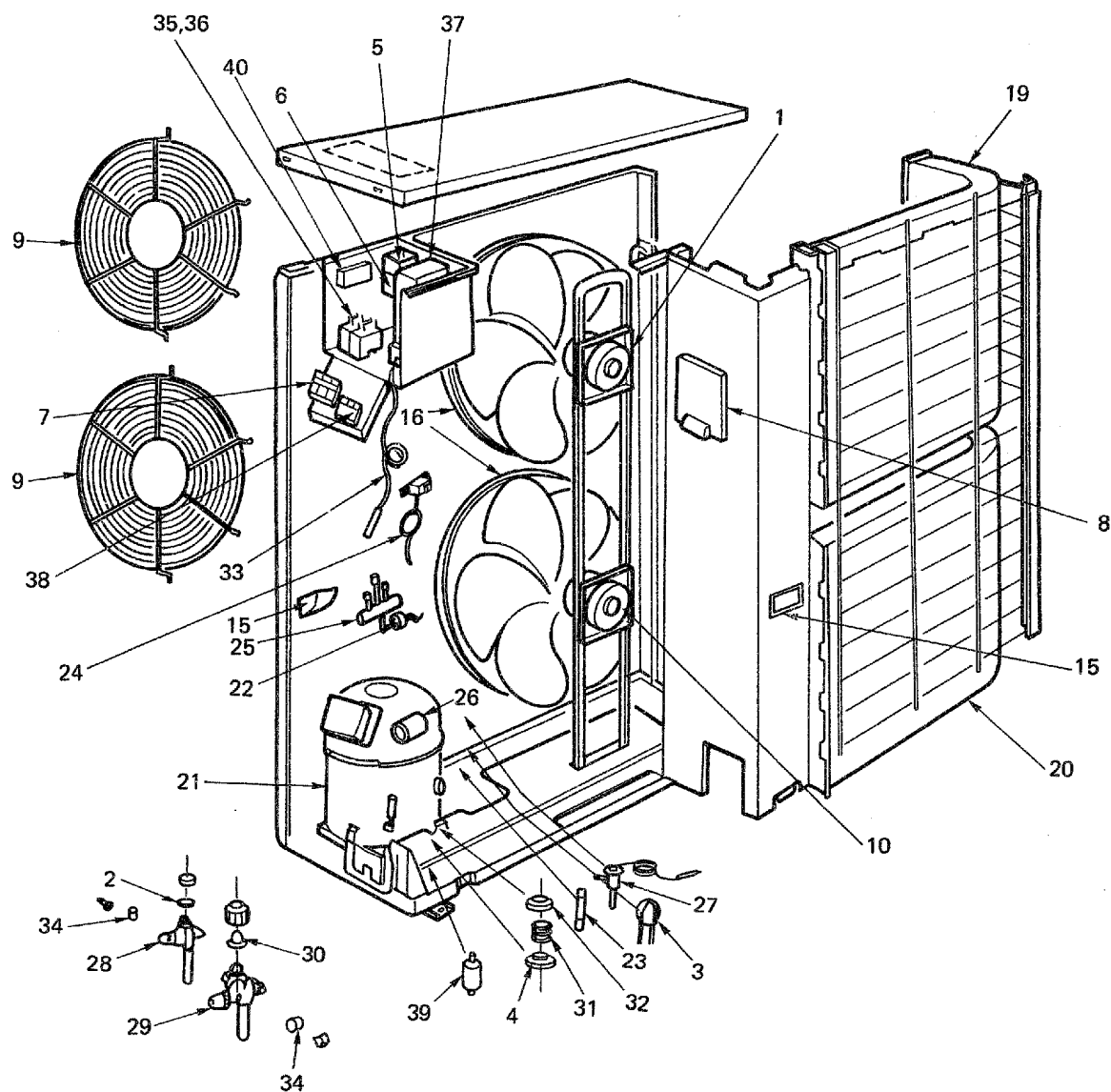
RAV-450CHJE/S450HJE
RAV-560CHJE/S560HJE
RAV-710CHE8/S710HE8
RAV-1000CHE8/S1000H
RAV-1250CHE8/S1250H

Location No.	Part No.	Description
1	43046137	Solenoid Coil
2	43046009	4-way Valve
3	43046114	Packed Valve, 1/4 in
4	43049132	Base, Spring, B
5	43191169	Guard, Fan
6	43153003	Relay, SA106H
9	43050244	Thermostat, Bimetal (RAV-S710HE8)
10	43169254	Holder, Thermostat (RAV-S710HE8)
11	43119368	Hanger
12	43120129	Fan, Propeller
13	43121273	Motor, AC 230V, 50Hz Fan (RAV-S450HJE, S560HJE)
13	43121287	Motor, AC 230V, 50Hz Fan (RAV-S710HE8)
14	43153371	Condenser (RAV-S450HJE)
14	43143369	Condenser (RAV-S560HJE)
14	43143361	Condenser (RAV-S710HE8)
15	43145082	Dryer
16	43146283	Checked Valve
17	43746867	Switch, High Pressure (RAV-S450HJE)
17	43146289	Switch, High Pressure (RAV-S560HJE, S710HE8)
18	43146328	Expansion Valve (RAV-S450HJE, S560HJE)
18	43146291	Expansion Valve (RAV-S710HE8)
19	43146347	Packed Valve, 1/2 in (RAV-S450HJE, S560HJE)
19	43146269	Packed Valve, 5/8 in (RAV-S710HE8)
20	43147195	Bonnet, 1/2 in (RAV-S450HJE, S560HJE)

Location No.	Part No.	Description
20	43194029	Bonnet, 5/8 in (RAV-S710HE8)
21	43147196	Bonnet
22	43149198	Spring, Buffer
23	43149212	Base, Spring, A
24	43154139	Switch, Magnetic (RAV-S450HJE, S560HJE)
24	43154140	Switch, Magnetic (RAV-S710HE8)
25	43155047	Capacitor, Electrolytic 4MFD, 350V AC (RAV-S450HJE, S560HJE)
25	43055274	Capacitor, Electrolytic 4MFD, 400V AC (RAV-S710HE8)
26	43160262	Terminal Block (RAV-S450HJE, S560HJE)
26	43160263	Terminal Block (RAV-S710HE8)
29	43162014	Cover
30	43169232	PC Board Assembly
31	43944047	Compressor, PH207JA-4 (RAV-S450HJE, S560HJE)
31	43944053	Compressor, YH277JA (RAV-S710HE8)
32	43049493	Block A (RAV-S450HJE)
34	43055304	Capacitor, Electrolytic 50MFD, 450V AC (RAV-S450HJE, S560HJE)
37	43150074	Defrost Thermostat (RAV-S450HJE, S560HJE)
37	43750836	Defrost Thermostat (RAV-S710HE8)

RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE8
 RAV-1250CHE8/S1250HE8

RAV-S1000HE8
 RAV-S1250HE8



RAV-450CHJE/S450HJE
 RAV-560CHJE/S560HJE
 RAV-710CHE8/S710HE8
 RAV-1000CHE8/S1000HE
 RAV-1250CHE8/S1250HE

Location No.	Part No.	Description
1	43121287	Motor AC230V, 50Hz Fan
2	43047401	Bonnet, 3/8 in
3	43047492	Tube, Capillary (1.0m) (RAV-S1000HE8)
3	43047527	Tube, Capillary (1.0m) (RAV-S1250HE8)
4	43049132	Base, Spring, B
5	43154131	Relay, SA205H
6	43153003	Relay, SA106H
7	43160264	Terminal Block
8	43062086	Cover
9	43191169	Guard, Fan
10	43121273	Motor, AC230V, 50Hz, Fan
15	43119368	Hanger
16	43120129	Fan, Propeller
19	43143392	Condenser, Upper (RAV-S1000HE8)
19	43143390	Condenser, Upper (RAV-S1250HE8)
20	43143391	Condenser, Lower (RAV-S1000HE8)
20	43143400	Condenser, Lower (RAV-S1250HE8)
21	43945029	Compressor, YH406JA (RAV-S1000HE8)
21	43946039	Compressor, YH506JA (RAV-S1250HE8)

Location No.	Part No.	Description
22	43046072	Solenoid Coil
23	43146283	Checked Valve
24	43146289	Switch, High Pressure
25	43146251	4-way Valve (RAV-S1000HE8)
25	43046008	4-way Valve (RAV-S1250HE8)
26	43149194	Block
27	43146362	Expansion Valve
28	43146219	Packed Valve, 3/8 in
29	43146200	Packed Valve, 3/4 in
30	43147451	Bonnet, 5/8 in
31	43149198	Spring, Buffer
32	43149212	Base, Spring, A
33	43150074	Defrost Thermostat
34	43147196	Bonnet, 1/4 in
35	43055274	Capacitor, Electrolytic 4MFD, 400V AC
36	43155047	Capacitor, Electrolytic 4MFD, 350V AC
37	43152232	Switch, Magnetic (RAV-S1000HE8)
37	43152255	Switch, Magnetic (RAV-S1250HE8)
38	43060336	Terminal Block, 6P
39	43145078	Dryer
40	43169268	PC Board Assembly

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